

Infosys Springboard Virtual Internship 7.0

Project Completion Report

Team Name: Team C | Batch 1 | Group 1

Team Members: Siddhant Singh, Priyam Sinha, Ponnuru Nithiin

Mentor Name: Ms. Nithyasri S J

Internship Duration: 8 weeks

Year: 2026

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Team Details

Batch Number: 1

Start date: 25/06/2026

Names: Siddhant Singh, Priyam Sinha, Ponnuru Nithiin

Internship Duration: 8 weeks

1. Project Title

Cyber Threat Analytics Platform with Incident Visualization System

2. Project Objective

The objective of this project is to design and deliver an interactive, end-to-end business-intelligence solution in Microsoft Power BI that consolidates large-scale cybersecurity incident data into a single, coherent analytical framework. The framework enables security analysts, decision-makers and communication teams to detect and quantify attack patterns, assess threat severity, monitor network and user behaviour, and convert 40,000 raw security events into clear, visually communicated and actionable intelligence that supports faster, evidence-based security decisions.

3. Project Description

■ 3.1 Overview

This project delivers an Advanced Threat Incident Analytics and Security Visualization Framework built entirely in Microsoft Power BI. It is based on a publicly available cybersecurity dataset of 40,000 incident records sourced from Kaggle, spanning January 2020 to October 2023 and covering 28 states across India. The raw data captures rich detail for every event - source and destination IP addresses and ports, protocol, packet type and length, traffic type, malware indicators, anomaly scores, attack type and signature, severity level, action taken, device, operating system, browser, network segment and geography.

The solution transforms this raw log data into seven themed, interconnected dashboards - Cyber Security Attacks Analysis, Threat Detection and Anomaly Analysis, Network Traffic Analysis, User Behavior Analytics, Threat Profile and Severity Breakdown, Security Alert Monitoring, and Attack Trends & Response Effectiveness. Together they provide a 360-degree view of the threat landscape, moving the reader from high-level KPIs down to granular, drill-able detail.

■ 3.2 Project Approach

The project was executed through a structured nine-step Power BI workflow. Each step is summarised below and elaborated in Section 5b.

- **Step 1 - Problem Definition:** framed the security questions and defined the KPIs and target audience.
- **Step 2 - Data Collection:** sourced a 40,000-record cybersecurity dataset from Kaggle.
- **Step 3 - Data Cleaning & Preprocessing:** shaped the data in Power Query (types, nulls, date/time split, category standardisation).
- **Step 4 - Data Modeling:** built a marked Date table and established relationships.
- **Step 5 - DAX Measures & Calculated Columns:** authored KPIs and the Anomaly Score Band logic.
- **Step 6 - Dashboard Design & Visualization:** designed seven themed report pages.

- **Step 7 - Interactivity & Advanced Analytics:** added slicers, cross-filtering, decomposition tree and drill-through.
- **Step 8 - Insight Generation & Interpretation:** derived and validated the findings.
- **Step 9 - Reporting, Documentation & Publishing:** produced the report, PPT, GitHub repository and mentor reviews.

3.3 Technology Used

The following tools and technologies were used across the internship:

- **Microsoft Power BI Desktop** - primary tool for data modeling, DAX and building all seven interactive dashboards.
- **Power Query Editor** - data cleaning, transformation and shaping (ETL): type conversion, null handling, date/time splitting and category standardisation.
- **DAX (Data Analysis Expressions)** - authoring measures and calculated columns such as Attack, High Severity, % High Severity, Malware Detected, Avg Anomaly Score, Block Rate % and Anomaly Score Band.
- **Microsoft Teams** - team collaboration, milestone meetings, screen-sharing and mentor reviews.
- **Microsoft PowerPoint** - milestone presentations and progress walkthroughs.
- **Microsoft Word** - project documentation and this milestone report.
- **GitHub** - version control and a central repository for the .pbix file and supporting artefacts.
- **Power BI Service (publishing)** - sharing dashboards for review and stakeholder access.

3.4 Insights from the Dashboard

Analysing the 40,000-record dataset across the seven dashboards surfaced the following key findings:

40,000 Total Attacks	33.5% High Severity	33.8% Block Rate	50% Malware (IoC)
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- The dataset spans 40,000 incidents across 2020–2023, with attack volume peaking in 2022 (10,750) and a lower 2023 figure (8,139) reflecting a partial year captured only to October.
- Attack types are almost evenly distributed, with DDoS marginally leading (33.6%), followed by Malware (33.3%) and Intrusion (33.2%) - indicating a broad, multi-vector threat environment rather than a single dominant attack.
- Severity is balanced across High (33.5%), Medium (33.6%) and Low (33.0%), meaning roughly one in three incidents is high-severity and warrants priority response.
- Only about 33.8% of attacks were actively Blocked, while the remainder were Ignored (33.2%) or merely Logged (33.0%) - a notable response gap and a clear improvement opportunity.
- Windows is by far the most-targeted operating system (~45% of incidents), followed by Linux (17.9%), Mac OS (14.5%), iOS (14.4%) and Android (8.2%).
- Mozilla accounts for roughly 80% of browser-associated activity, with Opera making up the remainder.
- The average anomaly score is 50.1 on a 0–100 scale, with incidents spread fairly evenly across all anomaly bands - the 51–75 band is the largest.
- Malware indicators (IoC Detected) are present in exactly half of all records (20,000), underscoring the prevalence of malware-linked activity.

- ICMP is the most common protocol (33.6%), and Control packets slightly outnumber Data packets (50.6% vs 49.4%).
- Incidents are widely distributed across 28 states - led by Manipur, Uttar Pradesh, Gujarat, Maharashtra and Arunachal Pradesh - showing a nationwide footprint rather than a few hotspots.

3.5 Real-World Impact for Media and Public Communication

Beyond the security operations centre, a framework of this kind has direct value for media and public communication. Cybersecurity data is often technical and intimidating; by distilling 40,000 raw events into clear KPIs, trends and geographic views, the dashboards make the threat landscape understandable to journalists, spokespeople, policymakers and the general public.

For communication teams, the visuals provide accurate, ready-to-use evidence for public advisories, awareness campaigns and executive briefings - for example, highlighting that roughly one in three incidents is high-severity, that malware indicators appear in half of all events, or that only a third of attacks are actively blocked. Presenting such figures visually builds credibility, counters misinformation, and helps citizens and organisations make informed decisions about their own digital safety. The state-level view further supports regionally targeted awareness messaging.

4. Timeline Overview

The eight-week internship progressed as follows, culminating in the Milestone 3 submission.

Week	Activities Planned	Activities Completed
Week 1	On boarding, problem understanding, dataset exploration	Kick off call; requirements defined; Kaggle dataset (40,000 records) finalised
Week 2	Data cleaning & pre-processing in Power Query	Cleaned data, corrected types, handled nulls, split date/time, standardised categories
Week 3	Data modelling & base DAX measures	Built & marked Date table, set relationships, created core measures (Attack, High Severity)
Week 4	First dashboards	Built Cyber Security Attacks & Network Traffic dashboards with KPI cards
Week 5	Anomaly & user-behaviour analytics	Anomaly Score Band, gauge, OS & browser analysis, user/device cards
Week 6	Two Final Dashboards	Security alert monitoring, Attack Trends & Response Effectiveness
Week 7	Testing validation and refinement	Refining final reports and documentation
Week 8	Documentation and Final presentation	Deploying report on gitlab and github

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Kick-off call completed; project requirements were defined and the 40,000-record Kaggle dataset was finalised.	25/06/2026
Prototype/First Draft	Initial dashboards were developed, including Cyber Security Attacks Analysis and Network Traffic Analysis with KPI cards.	16/07/2026
Mid-Term Review	Core analytical work was reviewed, including data modelling, DAX measures, anomaly analysis and user-behaviour dashboards.	29/07/2026
Final Submission	Final dashboards were validated and the report, PowerPoint presentation and GitHub repository were consolidated and prepared for submission.	19/08/2026
Presentation	Final project presentation and walkthrough were completed as part of Week 8 activities.	19/08/2026

The project was organised around three key milestones, each building on the previous one.

Milestone	Focus	Start Date
Milestone 1	Data Foundation & Understanding - dataset acquisition, cleaning, modelling and base KPIs	28/06/2026
Milestone 2	Core Dashboards & Behavioural Analytics - attack, traffic, anomaly and user-behaviour dashboards	16/07/2026
Milestone 3	Advanced Threat Profiling, Insights & Reporting - decomposition, severity profiling, insight synthesis and documentation	29/07/2026
Milestone 4	Security alert monitoring, Attack Trends & Response Effectiveness	05/08/2026

Milestone 1

Project Execution Details

1. Data Collection & Pre-processing (Milestone 1)

- Work completed during Week 1 covered the first stage of the pipeline acquiring the dataset and preparing it for analysis (Milestone 1). All preparation was carried out in Power Query, where every transformation was applied as a recorded, repeatable step. The query for this milestone contains 19 applied steps; the work they represent is summarised below in the order performed.
- Imported the raw data (Source). Connected Power Query to the "Cyber Security Attacks" dataset obtained from Kaggle 40,000 network-activity records described by 25 features and spanning January 2020 to October 2023 and loaded it into the editor as the starting point of the pipeline.
- Promoted the header row (Promoted Headers). Set the file's first row as column headers so that all 25 fields (Timestamp, Source and Destination IP Address, Protocol, Attack Type, Severity Level, Action Taken, and so on) were correctly named and addressable.
- Separated composite fields (Split Column by Delimiter, applied twice). Broke delimited columns into their individual components so that packed values could be analysed on their own - for example separating date

and time parts, and splitting combined values such as a geo-location or a device/browser string into distinct columns.

- Set and corrected data types (Changed Type, applied four times). Assigned the appropriate type to every column text, whole number, decimal and date/time and re-applied type changes after each structural change so the data stayed consistent; this is also where the Timestamp field was converted to a proper date/time type to enable later time-based analysis.
- Renamed columns for clarity (Renamed Columns, applied twice). Gave the newly derived and existing columns clear, consistent, analysis-friendly names.
- Cleaned and standardised values (Replaced Value, applied six times). Ran a series of find-and-replace operations to normalise the data replacing blank or placeholder entries, unifying inconsistent category labels, and correcting irregular values in fields such as Attack Type, Protocol, Severity Level and Action Taken so that each category is represented consistently.
- Removed redundant columns (Removed Columns). Dropped fields not required for the analysis (for example, raw columns already split into cleaner components and low-value fields such as payload text), reducing noise and file size.
- Removed duplicate records (Removed Duplicates). Eliminated repeated rows so that every security event is unique and event counts are not inflated.
- Removed error rows (Removed Errors). Discarded rows that produced conversion errors from invalid or malformed values, leaving only clean, well-typed records.

Together these steps produced a clean, consistent and fully analysis-ready dataset, which was then loaded through Close & Load for use in the next phase.

Milestone 2

Dashboard 1 - Cyber Security Attacks Analysis

The first dashboard is a single-screen overview of the entire attack landscape. A KPI band across the top gives the headline numbers, a geographic map anchors the layout, and a set of distribution and breakdown charts explain the composition of the attacks. Its visuals are:

4.1 KPI band

- **Total Attack** - a single-value card built on a distinct count of the Attack ID field, so it reports the full size of the analysed dataset - roughly 40,000 recorded attack events. Positioned at the top-left, it acts as the anchor KPI and establishes the denominator against which every other proportion on the page (attack-type shares, severity splits, protocol mix) should be read. Because it responds to selections made elsewhere on the page, clicking a segment in any chart updates this figure to show how many of the events fall inside the current selection.
- **Low Severity** - a KPI visual that isolates the number of events classified as Low severity, shown as a headline count with a compact trend indicator beneath it. Read next to the High Severity KPI, it gives an immediate sense of how the traffic splits between routine, low-risk activity and events that demand attention. A large

low-severity share signals that most recorded activity is background noise, which helps analysts calibrate how aggressively to triage.

- **High Severity** - the companion KPI to Low Severity, counting the events flagged as High severity and presenting them in the same headline-plus-trend format. This is the fastest read on the page for how much of what is being seen is serious, and the gap between this figure and the Low Severity count serves as a first, coarse prioritisation signal before the analyst drills into the charts below.

4.2 Distribution, breakdown and trend visuals

- **Geographic Spread (map)** - a bubble map that anchors the right-hand side of the dashboard and answers where the activity is coming from. Each bubble is positioned by geography (State, and at finer zoom City); bubble size is driven by the count of attack events, so high-volume locations dominate visually, while bubble colour segments each location by attack type (DDoS, Malware, Intrusion). A single glance therefore shows both how much activity a place generates and what kind. Occupying the full height of the layout, it forms the geographic backbone that the flat charts on the left complement.
- **Attacks by Attack Type (donut)** - a donut chart that breaks the total event count into the three attack categories - DDoS, Malware and Intrusion - with each arc sized by its count and the running total shown in the centre. Because the three categories come out broadly balanced in this dataset, the donut's core message is the absence of a single dominant threat: defensive effort has to be spread across all three rather than concentrated on one. The legend and data labels let the reader convert the arcs into approximate shares without leaving the visual.
- **Action Taken over Attacks (pie)** - a pie chart that slices the Total Attacks measure by the Action Taken field - the response each event received, such as Blocked, Logged or Ignored. Where the attack-type donut describes the threat, this describes the response, so together they show not only what is arriving but how the environment is reacting to it. A large Ignored or Logged-only slice relative to Blocked is worth attention, because it points to activity that passed through without active intervention.
- **Severity by Attack Type (stacked column)** - a stacked column chart with one column per attack type (DDoS, Malware, Intrusion), each column divided into Low, Medium and High severity segments and sized by event count. Reading across the columns shows whether the three attack types carry comparable total volume; reading the coloured segments within each column shows how the severity mix differs between them. Its value is prioritisation - an attack type with a visibly larger High-severity segment is a stronger candidate for fast response than one dominated by Low-severity events, even at similar totals. The Severity Sort helper column from Section 3 is what keeps Low, Medium and High in logical order here rather than alphabetical.
- **Traffic Breakdown (clustered bar)** - a clustered bar chart that groups the data by Traffic Type and, within each group, places a separate bar for each Protocol - TCP, UDP and ICMP - sized by event count. Clustering rather than stacking makes the protocols directly comparable inside every traffic category, so the reader can see at a glance which protocol carries the most traffic in each. This is operationally useful because monitoring and filtering rules are written per protocol: knowing whether attack traffic concentrates in TCP versus UDP or ICMP tells the team where packet inspection and rate-limiting will have the most effect.
- **Attacks over Time (line)** - a line chart running along the bottom of the dashboard that plots the Total Attacks measure against the Year level of the Date hierarchy built in Section 3, covering 2020 to 2023. It converts the otherwise static snapshot into a trend, answering whether the volume of recorded attacks is climbing, falling or holding steady year on year. Because it uses the same Date hierarchy that drives the slicers on Dashboard 2, an analyst can move from a year that looks unusual here to that same period on the anomaly dashboard to investigate why.

5. Dashboard 2 - Threat Detection and Anomaly Analysis

The second dashboard is an interactive investigation view. It centres on anomaly behaviour and lets the analyst slice the whole picture by date, geography, severity, protocol and attack type. A navigation button returns the user to the first dashboard, so the two operate as a linked pair. Its visuals are:

5.1 Analytical visuals

- **Anomaly Score Band (donut)** - the centrepiece of the dashboard: a donut that takes the Anomaly Score Band calculated column from Section 3 - which buckets the raw 0–100 anomaly scores into 0–25, 26–50, 51–75 and 76–100 - and sizes each arc by the summed anomaly score falling in that band. Turning a continuous score into four tiers lets an analyst read anomaly behaviour as clear severity bands rather than raw numbers, and the top band (76–100) is the one that matters most: it isolates the small pool of highly unusual, likely-suspicious events that warrant investigation first. Every slicer on the page reshapes this donut, so it also shows the anomaly profile of whatever subset the analyst has filtered to.
- **Average Anomaly Score (gauge)** - a gauge that summarises the average anomaly score across the currently selected events into a single needle position, giving a quick read of how unusual the slice is overall to sit alongside the banded donut. Where the donut shows the distribution across tiers, the gauge collapses it to one representative number, which is most useful in combination with the slicers: filtering to a particular city, protocol or attack type and watching the needle move shows whether that subset runs more or less anomalous than the baseline. A needle sitting high on the gauge is an early signal that the current selection is dominated by out-of-pattern activity.
- **Attacks by Browser (bar chart)** - a bar chart that ranks the Browser (user-agent) values - for example Mozilla, Opera and Chrome - by the Total Attacks measure, with one bar per browser so the longest bar names the most-seen client. Presenting it as bars rather than a donut makes the ranking explicit and the differences between clients easy to compare directly. It adds client-side context to an investigation: once the anomaly bands and slicers have narrowed the picture, the browser breakdown helps characterise what kind of client the suspicious activity is coming through, which can support attribution and pattern-matching.
- **Attacks by State (area)** - a wide area chart along the foot of the dashboard that plots the Total Attacks measure across States, the filled area emphasising where volume rises and falls from one state to the next. It tells the same geographic story as the bubble map on Dashboard 1 but in a flat, easy-to-scan form that suits an analytical page and updates instantly as slicers are applied. Read together with the slicer panel, it lets the analyst confirm whether a filtered threat scenario is geographically concentrated or spread evenly across states.

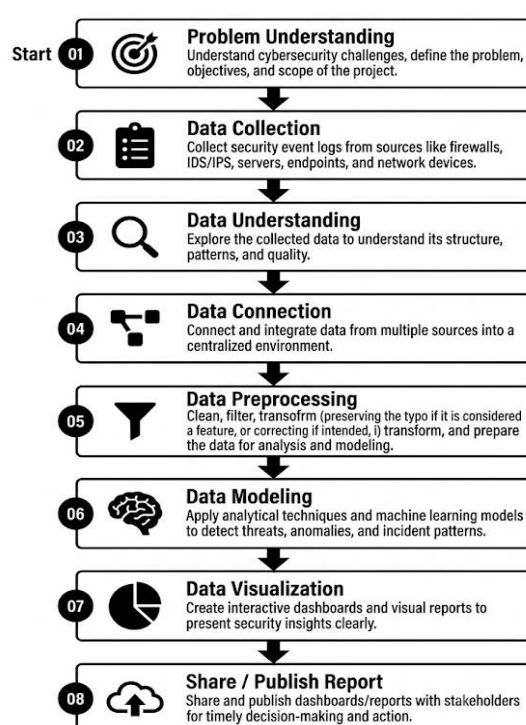
5.2 Interactivity

- **Slicer panel** - a column of five slicers down the left edge - Date, City, Severity Level, Protocol and Attack Type - that filter every visual on the page at once. Because they act together, an analyst can compose a precise scenario (for example, High-severity Intrusion events on TCP in a chosen city over a set date range) and watch the donut, gauge, bar and area charts all recompute to describe just that subset. The Date slicer draws on the same Date hierarchy used by the time-trend line on Dashboard 1, and the City slicer pairs with the State area chart to move between fine and coarse geography.
- **Cross-filtering** - beyond the slicers, the visuals cross-filter one another: clicking a segment in one chart filters the rest of the page to that selection. Selecting the 76–100 arc on the anomaly donut, a specific browser bar, or a state in the area chart immediately narrows every other visual to matching events, so an investigation can proceed by clicking rather than by manually setting slicers. This turns the page into an exploratory surface where each question - for instance, what the highly-anomalous slice looks like by browser and location - is one click away from an answer.

- **Back navigation button** - an in-report button, placed at the top-left of the page, that navigates back to the Cyber Security Attacks Analysis dashboard. It makes the two pages behave as one connected experience rather than two separate files: an analyst who spots something in the high-level overview can jump into the anomaly explorer to dig in, then return to the overview without leaving the report. This linked-pair design is what lets the descriptive and investigative views operate as a single tool.

5b. Project Execution Details

This section details how the framework was executed end-to-end. The project followed a structured nine-stage Power BI workflow, after which each of the seven dashboards was assembled, formatted and validated. The workflow below summarises the pipeline from problem definition to publishing.



Execution Workflow

Figure 1 - End-to-end Power BI execution workflow (nine stages).

Step 1 - Problem Definition & Requirement Understanding. We began by framing the core security questions the framework needed to answer - how many attacks occur, of what type and severity, how the network and users behave, and how effectively threats are handled. From these questions we defined the KPIs (total attacks, high-severity share, block rate, average anomaly score, malware detections) and identified the target audience.

Step 2 - Data Collection. A publicly available cybersecurity dataset of 40,000 incident records was sourced from Kaggle. It was chosen for its breadth - over thirty attributes per event covering network, threat, user and geographic dimensions - making it well-suited to a multi-dashboard analytical framework.

Step 3 - Data Cleaning & Preprocessing. Using the Power Query Editor, the raw data was cleaned and shaped through nineteen individual transformation steps, grouped below into eleven logical stages, so that the dataset was fully analysis-ready before modeling began.

- 1. Removed Duplicate Records** - removed duplicate rows to eliminate redundant data and improve data accuracy.
- 2. Changed Timestamp Data Type** - converted the Timestamp column from Text to Date/Time format for time-based analysis.
- 3. Created Date, Hour & Day Columns** - extracted Date, Hour, and Day Name from the Timestamp column for trend analysis.
- 4. Handled Missing Values** - replaced null values in Malware Indicators, Alerts/Warnings, Proxy Information, Firewall Logs, and IDS/IPS Alerts with "No".
- 5. Split Geo-location Column** - split the Geo-location column into two separate columns, City and State, using a comma delimiter.
- 6. Renamed Columns** - renamed the newly created columns to City and State for better readability and consistency.
- 7. Extracted Browser Information** - created a new Browser column by extracting browser names from the Device Information column using conditional logic.
- 8. Removed Payload Data Column** - deleted the Payload Data column because it contained unnecessary placeholder text and did not contribute to analysis.
- 9. Created Severity Sort Key** - added a Severity Sort column (Low = 1, Medium = 2, High = 3) to ensure the severity levels are displayed in the correct order.
- 10. Changed Data Types** - assigned appropriate data types to the transformed columns (Date/Time, Whole Number, Text, etc.) to improve data consistency.
- 11. Created Anomaly Score Bands** - grouped Anomaly Score values into ranges (0–25, 25–50, 50–75, 75–100) for easier visualization and analysis.

Step 4 - Data Modeling. A dedicated Date table was created and marked as the model's date table to enable time-based analysis, and relationships were established so that all visuals respond consistently to slicers and cross-filtering.

Step 5 - DAX Measures & Calculated Columns. Core DAX logic was authored, including Attack = COUNTROWS('cybersecurity_attacks'), High Severity and % High Severity, Malware Detected, Avg Anomaly Score, Block Rate %, and the Anomaly Score Band calculated column that buckets scores into 0–25, 26–50, 51–75 and 76–100.

Step 6 - Dashboard Design & Visualization. Seven themed report pages were designed with a consistent dark security aesthetic, KPI cards and a clear visual hierarchy, so each page tells a focused story.

Step 7 - Interactivity & Advanced Analytics. Slicers (date, severity, attack type), cross-highlighting, tooltips and advanced visuals - the decomposition tree, ribbon charts, treemaps and matrices - were added so users can explore the data interactively and drill from summary to detail.

Step 8 - Insight Generation & Interpretation. The completed dashboards were analysed to extract the findings documented in Section 3.4, validating figures across visuals for consistency.

Step 9 - Reporting, Documentation & Publishing. Findings were consolidated into this Word report and a PowerPoint walkthrough; the .pbix file and artefacts were version-controlled on GitHub; progress was reviewed with the mentor on Microsoft Teams; and the report was prepared for the Milestone 4 submission.

Dashboard 1 - Cyber Security Attacks Analysis

Key Performance Indicators (KPIs)

Located at the very top, these three text boxes provide a high-level summary of the attack volume:

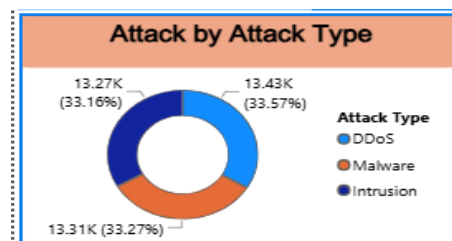
- **Total Attack:** Shows a total of **40K** recorded cybersecurity incidents.
- **High Severity:** Indicates that **13382** of those attacks were classified as having a high severity level.
- **Low Severity:** Shows that **13183** of the attacks were classified as low severity. *(The remaining balance accounts for medium-severity attacks).*

Cyber Security Attacks Analysis		
Total Attack	High Severity	Low Severity
40K	13382	13183

Attack by Attack Type

This donut chart on the middle-left breaks down the total attacks into three specific categories, showing a nearly equal distribution among them:

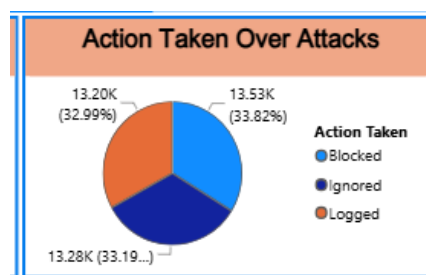
- **DDoS:** 13.43K (33.57%)
- **Malware:** 13.31K (33.27%)
- **Intrusion:** 13.27K (33.16%)



Action Taken Over Attacks

Located in the center, this pie chart illustrates the system's or security team's response to the detected incidents. Similar to the attack types, the actions are evenly distributed:

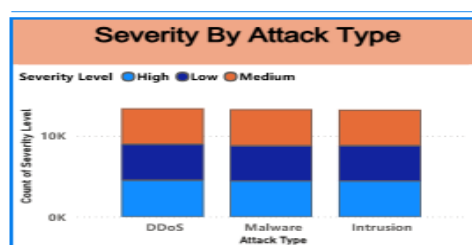
- **Blocked:** 13.53K (33.82%)
- **Logged:** 13.28K (33.19%)
- **Ignored:** 13.20K (32.99%)



Severity By Attack Type

This stacked column chart in the bottom-left corner provides a deeper look into the attack types (DDoS, Malware, Intrusion) by segmenting each column into three severity levels:

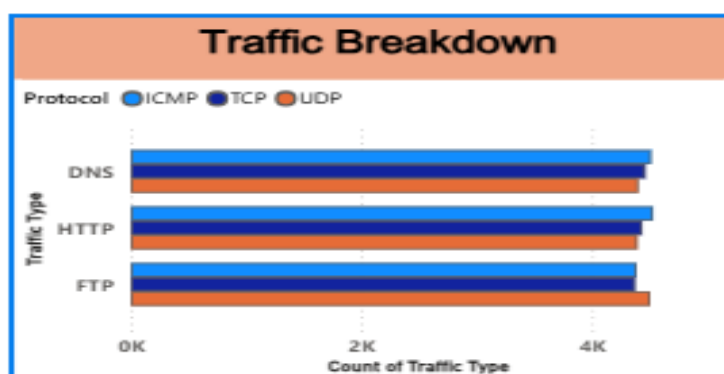
- **High** (Light Blue)
- **Low** (Dark Blue)
- **Medium** (Orange) The chart reveals that the proportion of high, low, and medium severity incidents is highly consistent regardless of the attack type.



Traffic Breakdown

Found in the bottom-middle section, this clustered horizontal bar chart categorizes network traffic involved in the attacks:

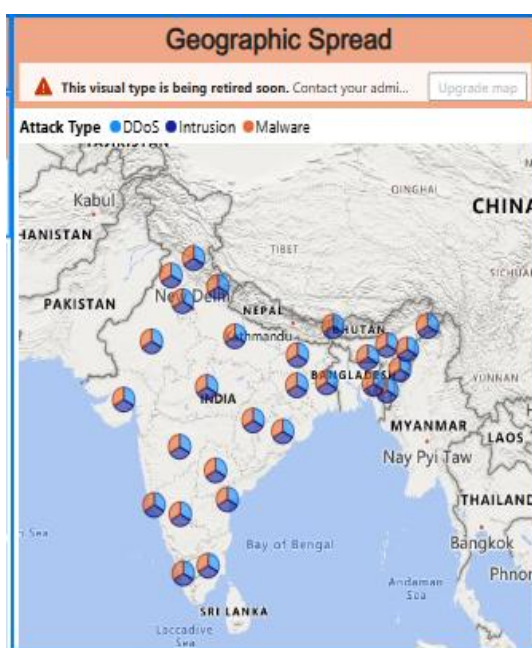
- **Traffic Types:** DNS, HTTP, and FTP.
- **Protocols:** Each traffic type is further broken down by ICMP (Light Blue), TCP (Dark Blue), and UDP (Orange).
- The data displays a uniform distribution, with each protocol type accounting for roughly 4K incidents across all three traffic categories.



Geographic Spread

Taking up the top right section, this map visualizes the physical origins or targeted nodes of the attacks:

- The map focuses on **India and neighboring countries**.
- It uses multi-colored circular markers at various geographic points. The colors within these markers correspond to the **Attack Type** (DDoS, Intrusion, Malware), indicating the mix of attacks hitting each specific location.
- *Note:* There is an administrative warning banner at the top of this chart stating, "This visual type is being retired soon."



Total Attack by Year

Located in the bottom right, this line graph tracks the volume of attacks over a four-year timeline:

- **2020 to 2022:** The number of attacks remains stable, tracking steadily just above the 10K mark.
- **2023:** The trend line shows a sharp, noticeable decline in total attacks dropping below the 8K mark.



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This landing dashboard gives a portfolio-level overview of the entire threat environment. A Total Attack card headlines the volume at 40,000 events, flanked by High Severity and Low Severity KPI indicators. The Total Attacks by Year line chart tracks volume over 2020–2023, revealing a peak in 2022 and a tapering 2023 (partial year). The Severity by Attack Type clustered column chart compares High, Medium and Low counts across DDoS, Intrusion and Malware, while the Attack Type donut shows their near-equal split. The Action Taken over Attacks pie exposes the response mix (Blocked / Ignored / Logged), the Traffic Breakdown bar chart cross-tabulates Protocol against Traffic Type (DNS / HTTP / FTP), and the Geographic Spread map plots attack types across Indian states.

Dashboard 2 - Threat Detection and Anomaly Analysis

Dashboard Filters

Located at the top left, these three dropdowns and date pickers indicate that the data currently displayed on the dashboard is filtered for specific criteria:

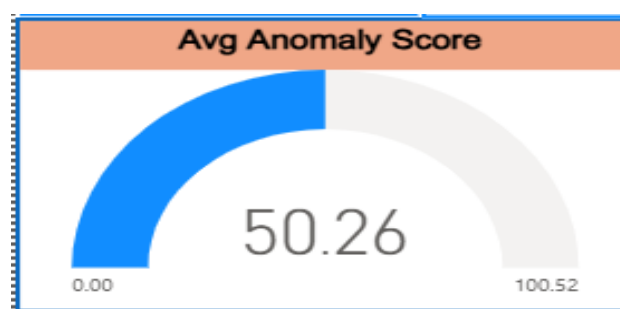
- **Severity Level:** Currently set to filter for **High** severity incidents.
- **Attack type:** Currently set to filter for **DDoS** attacks.
- **Date:** The date range is restricted from **03-01-2020** to **05-10-2023**.

Severity Level	Attack type	Date
High	DDoS	03-01-2020 05-10-2023

Avg Anomaly Score

This gauge chart on the middle-left displays the average anomaly score based on the applied filters:

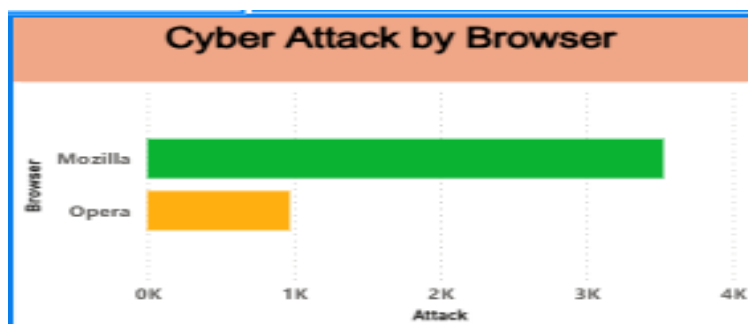
- **Score:** The central number indicates an average score of **50.26**.
- **Scale:** The gauge ranges from a minimum of **0.00** to a maximum of **100.52**. The blue fill represents the score sitting precisely at the halfway point of the scale.



Cyber Attack by Browser

Positioned in the middle center, this horizontal bar chart compares the volume of attacks originating from or targeting specific web browsers:

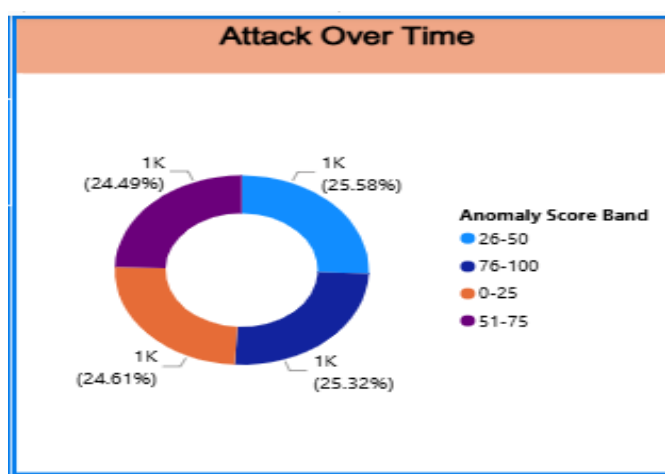
- **Mozilla:** The green bar shows a significantly higher attack count, reaching approximately **3.5K**.
- **Opera:** The yellow bar indicates a lower attack count, sitting right at the **1K** mark.



Attack Over Time (Anomaly Score Band)

Located on the right side, there is a notable discrepancy in this visual: the title says "Attack Over Time," but the visual is actually a donut chart representing **Anomaly Score Bands**. It shows an almost perfectly even distribution across four quartiles:

- **26-50** (Light Blue): 1K (25.58%)
- **76-100** (Dark Blue): 1K (25.32%)
- **0-25** (Orange): 1K (24.61%)
- **51-75** (Purple): 1K (24.49%)

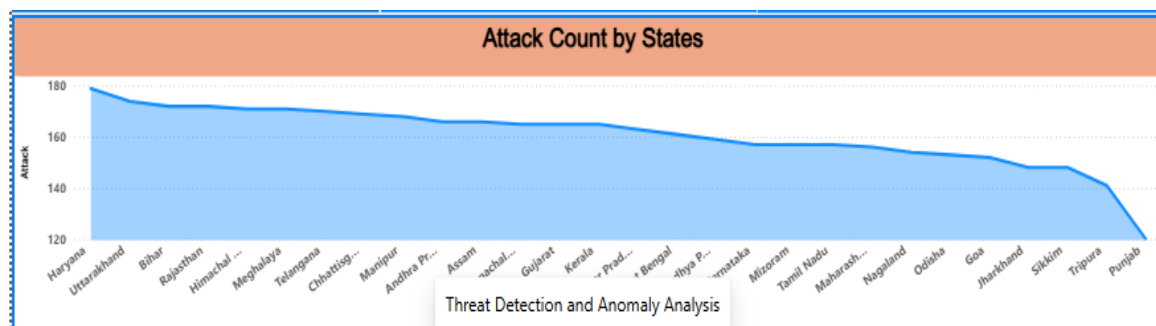


Attack Count by States

Taking up the entire bottom section, this filled area chart visualizes the distribution of attacks across various Indian states:

- **X-Axis:** Lists numerous Indian states, starting with Haryana on the left and ending with Punjab on the right.

- **Y-Axis:** Represents the count of attacks, scaled from 120 up to 180.
- **Trend:** The chart displays a smooth, gradual downward slope. Haryana records the highest number of attacks (approaching 180), while the volume steadily decreases across the listed states, with Punjab recording the lowest count (dropping down to 120).



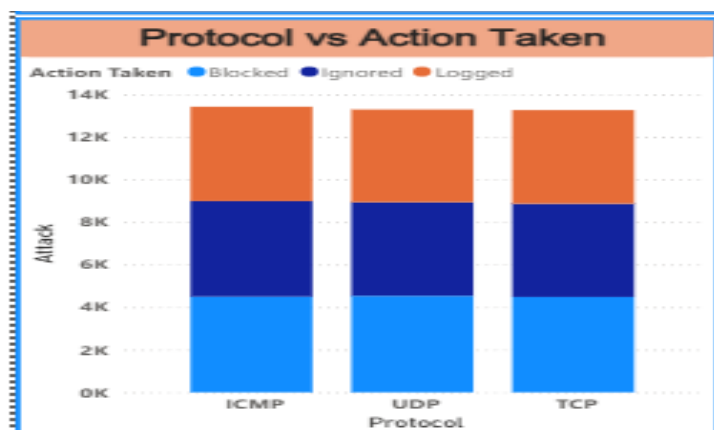
This page focuses on anomaly-based detection and is driven by Date, Severity Level and Attack Type slicers for interactive filtering. The Avg Anomaly Score gauge summarises detection risk at 50.1 out of 100. The Attacks by Anomaly Band donut distributes events across the four score bands; the Cyber Attacks by Browser bar chart contrasts Mozilla and Opera exposure; and the Attack Count by State area/column chart shows the geographic concentration of incidents. Together these visuals help analysts isolate high-anomaly, high-severity segments quickly.

■ Dashboard 3 - Network Traffic Analysis

Protocol vs Action Taken

Located in the top-left corner, this stacked column chart illustrates how the security system responded to attacks across different network protocols:

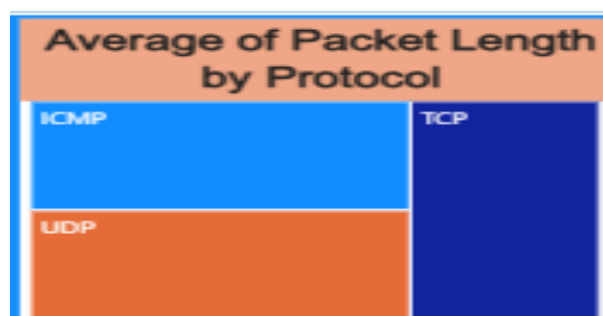
- **X-Axis (Protocol):** ICMP, UDP, and TCP.
- **Y-Axis (Attack):** Measures the volume of attacks, reaching just under 14K for each column.
- **Categories:** Each protocol's column is evenly split into three actions taken: **Blocked** (Light Blue), **Ignored** (Dark Blue), and **Logged** (Orange).
- **Trend:** The distribution of attacks and the subsequent actions taken are virtually identical across all three protocols.



Average of Packet Length by Protocol

Positioned in the top-middle, this treemap visualizes the proportion of the average packet length associated with each protocol:

- **TCP** (Dark Blue) occupies the entire right half of the visual, indicating a significant portion of the average packet length.
- **ICMP** (Light Blue) and **UDP** (Orange) share the left half, with ICMP situated at the top and UDP at the bottom.



Protocol vs Severity

Found directly below the treemap, this table acts as a heatmap detailing the exact count of attacks by protocol and severity level:

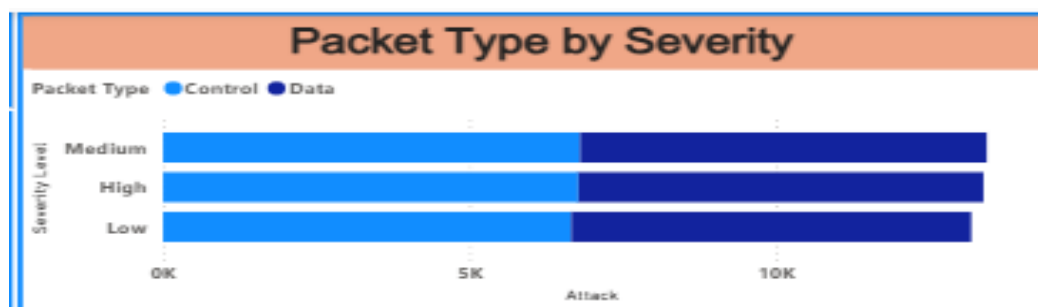
- **Rows:** ICMP, TCP, UDP.
- **Columns:** High (Red), Low (Green), Medium (Yellow).
- **Data:** The attack counts are highly uniform across all intersections, ranging narrowly from 4292 (UDP - Low) to 4552 (UDP - Medium), indicating no single protocol is disproportionately favored for high-severity attacks.

Protocol vs Severity			
Protocol	High	Low	Medium
ICMP	4455	4505	4469
TCP	4472	4386	4414
UDP	4455	4292	4552

Packet Type by Severity

Located in the top-right, this stacked horizontal bar chart breaks down severity levels by the type of network packet involved:

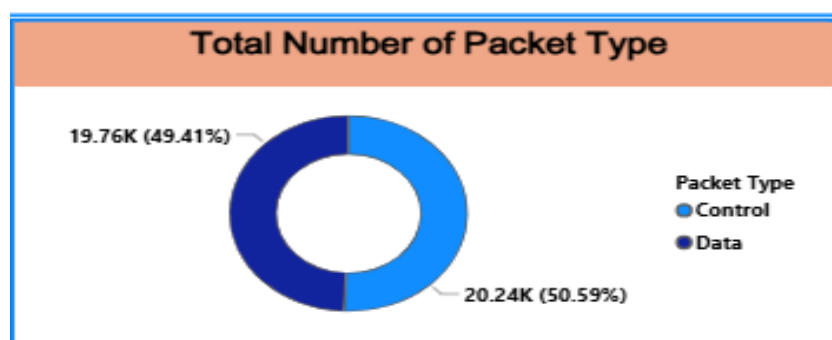
- **Y-Axis:** Severity Level (Medium, High, Low).
- **X-Axis:** Total attack count, reaching just above 10K per severity level.
- **Categories:** Each severity bar is split into **Control** packets (Light Blue) and **Data** packets (Dark Blue). The split is perfectly even (approximately 50/50) across all three severity levels.



Total Number of Packet Type

Positioned in the middle-right, this donut chart provides a straightforward summary of the overall packet types involved in the recorded attacks:

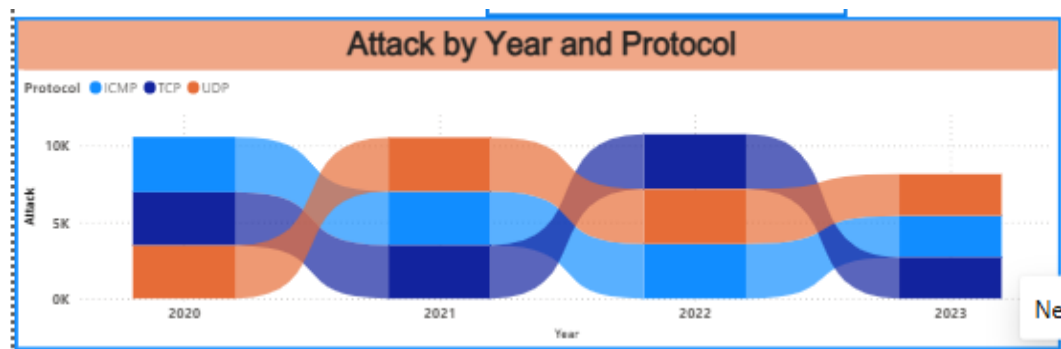
- **Control** (Light Blue): **20.24K** packets, representing **50.59%** of the total.
- **Data** (Dark Blue): **19.76K** packets, representing **49.41%** of the total.
- This confirms an almost equal distribution between control and data packets overall.



Attack by Year and Protocol

Taking up the bottom-left section, this ribbon chart (or smoothed stacked area chart) illustrates how the volume and composition of attacks have shifted over time:

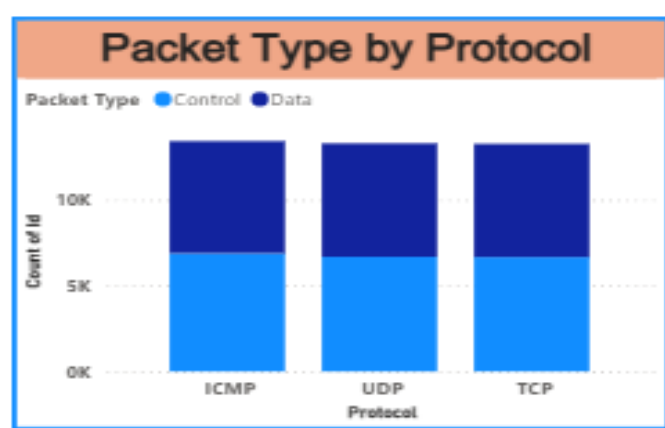
- **X-Axis:** Years 2020 to 2023.
- **Y-Axis:** Attack volume (0K to slightly above 10K).
- **Trend:** While the overall volume stays relatively stable from 2020 to 2022, the prominence of each protocol (ICMP in Light Blue, TCP in Dark Blue, UDP in Orange) fluctuates. For example, ICMP drops in rank during 2021, while UDP rises. Consistent with other dashboard views, there is a distinct drop in overall attack volume in **2023**.



Packet Type by Protocol

Located in the bottom-right corner, this stacked column chart shows the composition of packet types within each specific protocol:

- **X-Axis (Protocol):** ICMP, UDP, TCP.
- **Y-Axis (Count of Id):** Total counts reaching roughly 13K per protocol.
- **Categories:** Just like the "Packet Type by Severity" visual, every column here is divided into a near-perfect 50/50 split between **Control** (Light Blue) and **Data** (Dark Blue) packets, showing high uniformity across the network protocols.



This dashboard dissects network-level behaviour. Packet Type by Protocol and Protocol vs Action Taken columns show how Control/Data packets and response actions distribute across ICMP, TCP and UDP. Packet Type by Severity relates packet composition to threat severity, while the Protocol vs Severity matrix provides exact counts. A Total Packet Type donut summarises the Control-versus-Data split (50.6% vs 49.4%), a ribbon chart ranks protocols by attack volume, and an Avg Packet Length by Protocol view profiles packet size - giving a complete picture of traffic characteristics.

Dashboard 4 - User Behaviour Analytics

Monthly Attack matrix

Located in the top-left corner, this table acts as a heatmap showing the volume of attacks broken down by month and attack type:

- **Rows & Columns:** The rows display the months of the year (January to December), and the columns are split by attack type (DDoS, Intrusion, Malware).
- **Color Gradient:** The cells are color-coded from green (lowest volume) to red (highest volume).
- **Trend:** The highest concentration of attacks (in red and orange) occurs in the spring and summer months, notably March and June. The lowest volume of attacks (in bright green) consistently appears at the end of the year in November and December, dropping into the 800-900 range.

Key Performance Indicators (KPIs)

Located top-center, these two summary cards display overall user and device metrics:

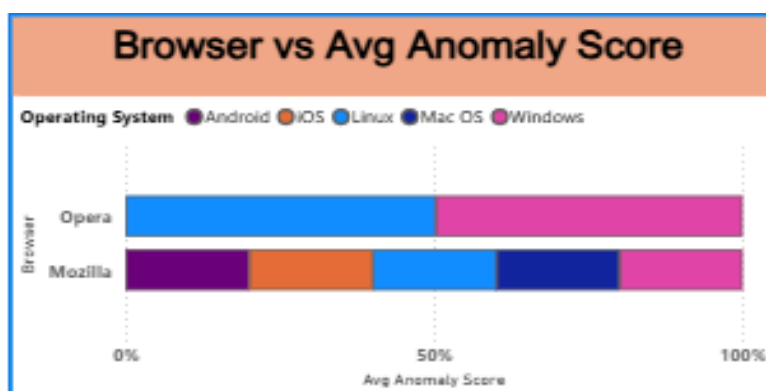
- **Total Users:** 32.389K
- **Total devices:** 32.104K
- This indicates a nearly 1:1 ratio of users to devices being monitored within this dataset.

Monthly Attack matrix			
Month	DDoS	Intrusion	Malware
January	1099	1116	1163
February	1085	1107	1040
March	1299	1182	1197
April	1166	1140	1115
May	1200	1212	1183
June	1190	1268	1151
July	1163	1224	1236
August	1226	1173	1216
September	1221	1089	1172
October	1015	961	1013
November	869	899	935
December	895	894	886

Browser vs Avg Anomaly Score

Positioned in the middle of the dashboard, this 100% stacked horizontal bar chart illustrates how anomaly scores are distributed across operating systems for two specific browsers:

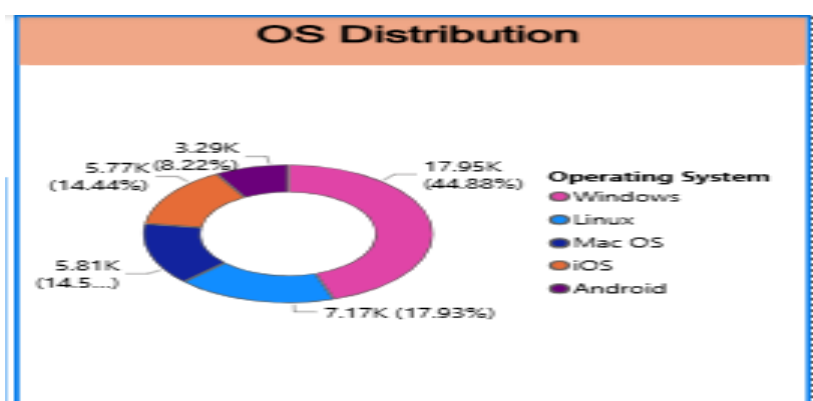
- **Opera:** The anomaly score distribution is split cleanly and evenly between just two operating systems: Linux (Light Blue) and Windows (Pink).
- **Mozilla:** The anomaly score distribution is spread across all five tracked operating systems (Android, iOS, Linux, Mac OS, Windows) in relatively even proportions.



OS Distribution

Found in the top-right corner, this donut chart breaks down the total count of operating systems involved in the data:

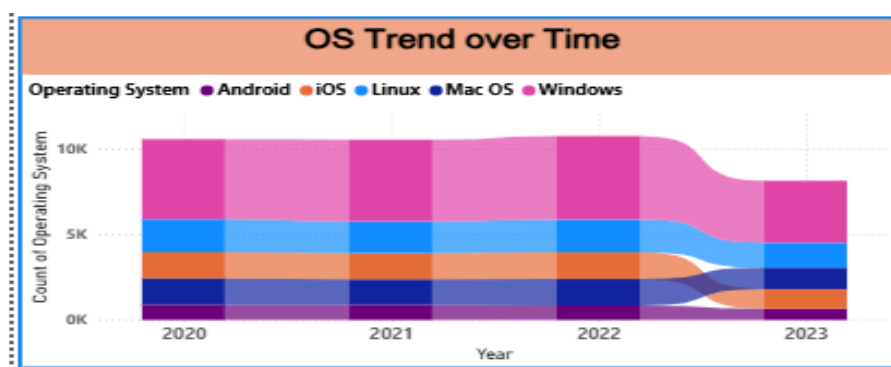
- **Windows (Pink):** Represents the largest segment by a significant margin at **17.95K (44.88%)**.
- **Linux (Light Blue):** The second largest at **7.17K (17.93%)**.
- **Mac OS (Dark Blue):** Accounts for **5.81K (~14.5%)**.
- **iOS (Orange):** Accounts for **5.77K (14.44%)**.
- **Android (Purple):** The smallest segment at **3.29K (8.22%)**.



OS Trend over Time

Located in the bottom-left, this stacked area chart tracks the volume of operating systems recorded from 2020 to 2023:

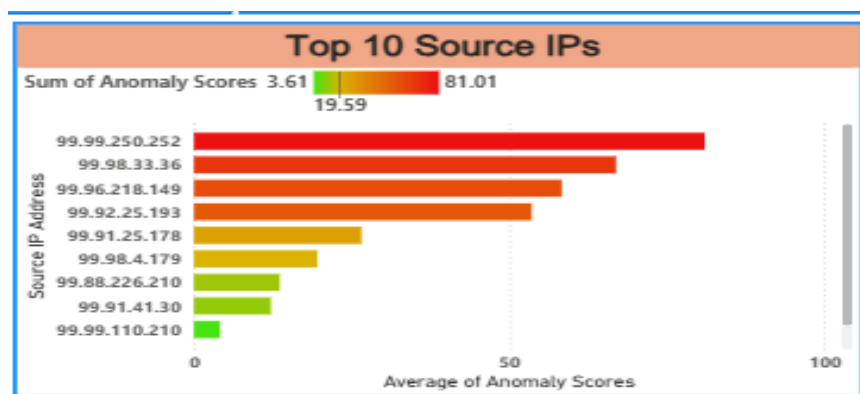
- **Distribution:** It visually mirrors the "OS Distribution" donut chart, with Windows (Pink) taking up the largest top band, followed by the other operating systems stacked below it.
- **Timeline Trend:** The total volume remains stable and flat from 2020 through 2022. Consistent with the other dashboard tabs, there is a sharp and noticeable decline in overall volume in **2023**.



Top 10 Source IPs

Taking up the bottom-right section, this horizontal bar chart ranks the specific IP addresses responsible for the highest anomaly scores:

- **X-Axis:** Measures the Average of Anomaly Scores (0 to 100).
- **Color Gradient:** The bars are colored based on the *Sum* of Anomaly Scores, ranging from Green (3.61) to Red (81.01).
- **Top Threat:** The top IP address (99.99.250.252) stands out with a significantly higher average anomaly score (and a deep red color) compared to the rest of the list. As you move down the top 10, the bars get progressively shorter and shift toward yellow and green, indicating lower anomaly severity.



This page profiles the human and device dimension. Total Users (32,389) and Total Devices (32,104) cards quantify the affected population. The OS Distribution donut shows Windows dominating at roughly 45%; the Browser vs Avg Anomaly Score visual compares risk across browsers and operating systems; and the OS Trend over time and Monthly Attack Matrix reveal temporal patterns by attack type. The Top 10 Source IPs chart surfaces the highest-anomaly source addresses for investigation.

■ Dashboard 5 - Threat Profile and Severity Breakdown

Attacks by Attack Type x Severity

Located in the top-left corner, this cross-tabulation table provides a precise numerical breakdown of attacks:

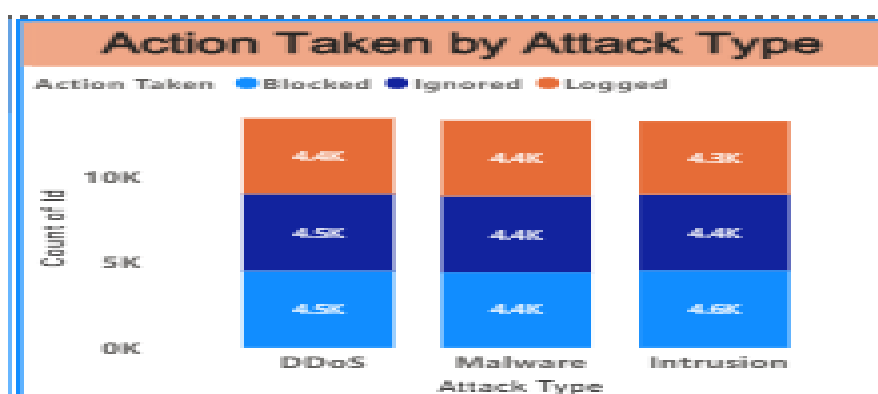
- **Rows:** Represents the Attack Type (DDoS, Intrusion, Malware).
- **Columns:** Represents the Severity Level (High, Low, Medium).
- **Data:** It displays the exact count of incidents for each intersection, culminating in total counts. The grand total of all attacks is exactly **40000**. The data shows a remarkably even spread, though DDoS has a marginally higher total (13428) compared to the others.

Attack Type	High	Low	Medium	Total
DDoS	4523	4450	4455	13428
Intrusion	4427	4374	4464	13265
Malware	4432	4359	4516	13307
Total	13382	13183	13435	40000

Action Taken by Attack Type

Positioned in the top-middle, this stacked column chart illustrates the security system's response to different threats:

- **X-Axis:** Attack Type (DDoS, Malware, Intrusion).
- **Y-Axis:** Count of Id (Attack volume).
- **Categories:** Each column is segmented into the action taken: **Blocked** (Light Blue), **Ignored** (Dark Blue), and **Logged** (Orange).
- **Trend:** The distribution of the actions taken is highly uniform across all three attack types, with each action segment accounting for roughly 4.3K to 4.6K incidents.



Key Performance Indicators (KPIs)

Found in the top-right corner, these two standalone cards highlight specific top-level metrics:

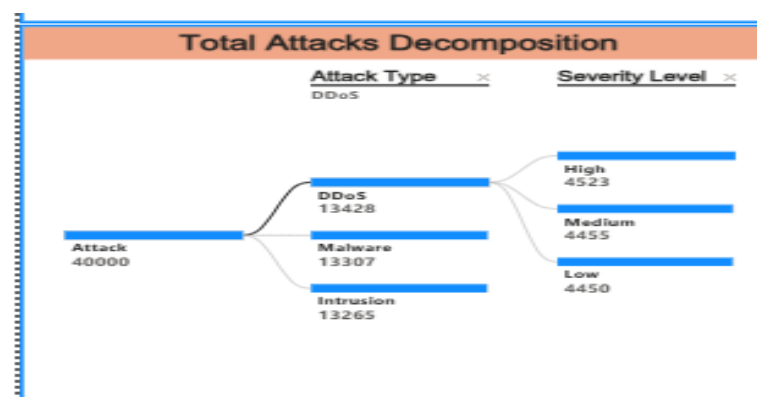
- **% High Severity:** Displays a value of **0.33** (representing 33%), indicating the proportion of the total attacks that were classified with a high severity level.
- **Malware Detected:** Displays a count of **20K**, indicating the total volume of malware incidents detected in this specific context.



Total Attacks Decomposition

Taking up the bottom-left section, this decomposition tree visualizes the hierarchical breakdown of the attack data:

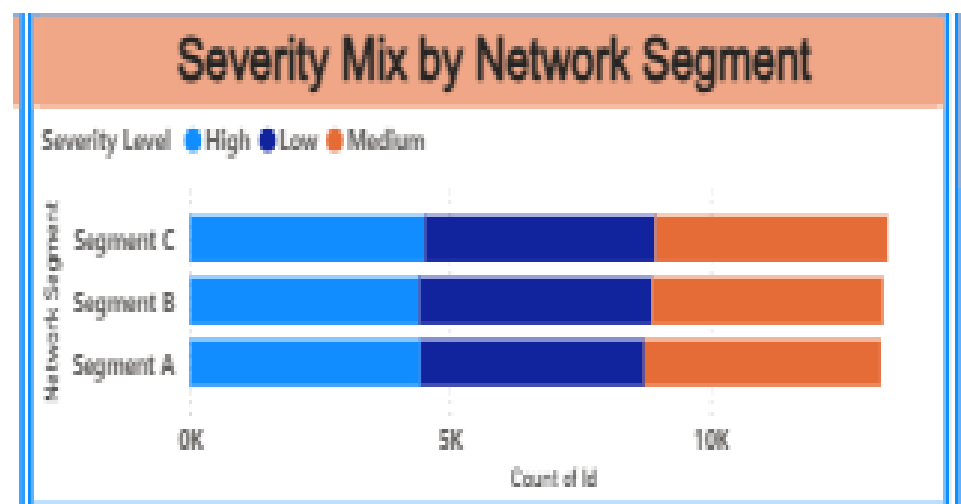
- **Root:** Starts with the total attack volume of **40000**.
- **First Branch (Attack Type):** Splits the total into DDoS (13428), Malware (13307), and Intrusion (13265).
- **Second Branch (Severity Level):** The visual specifically expands the DDoS branch to show its composition by severity: High (4523), Medium (4455), and Low (4450).



Severity Mix by Network Segment

Located in the middle-right area, this stacked horizontal bar chart shows how attack severities are distributed across different parts of the network:

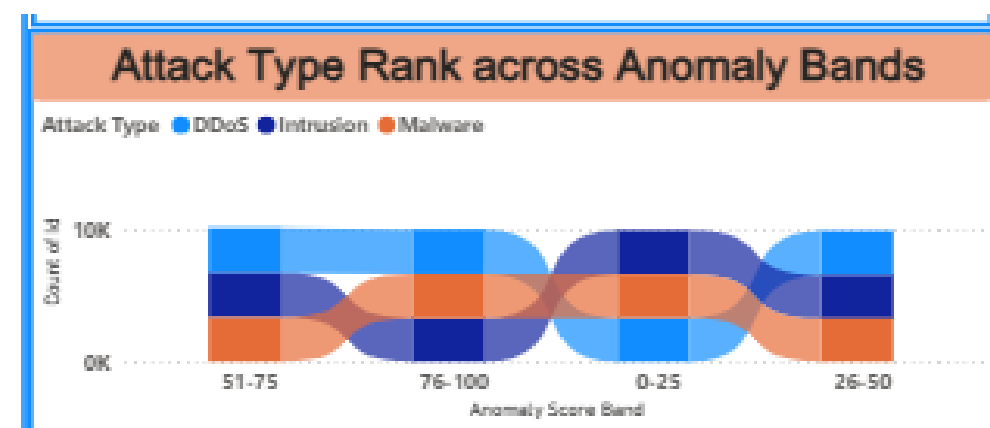
- **Y-Axis:** Network Segments (Segment C, Segment B, Segment A).
- **X-Axis:** Count of Id.
- **Categories:** High (Light Blue), Low (Dark Blue), and Medium (Orange) severity levels.
- **Trend:** The chart reveals that every network segment experiences a virtually identical volume and proportional mix of high, low, and medium severity attacks.



Attack Type Rank across Anomaly Bands

Positioned in the bottom-middle, this ribbon chart tracks the flow and ranking of attack types across different scoring buckets:

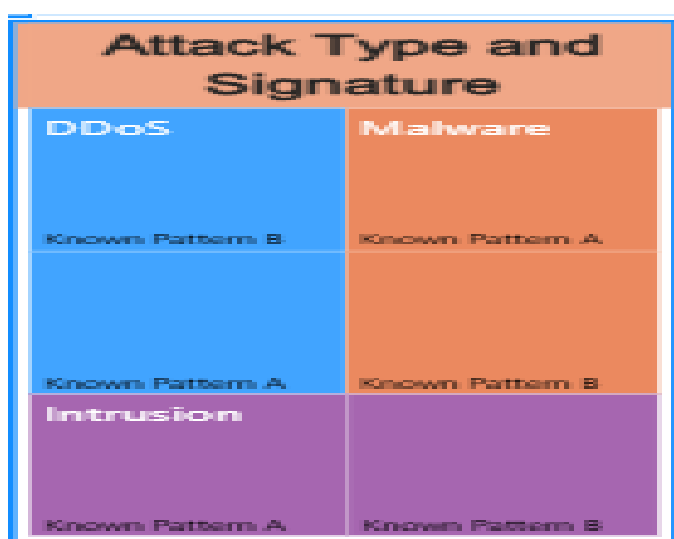
- **X-Axis:** Anomaly Score Bands (non-sequential: 51-75, 76-100, 0-25, 26-50).
- **Categories:** DDoS (Light Blue), Intrusion (Dark Blue), and Malware (Orange).
- **Trend:** The twisting ribbons illustrate how the relative rank and volume of each attack type fluctuate depending on the specific anomaly score band they fall into.



Attack Type and Signature

Located in the far bottom-right corner, this treemap visualizes the composition of specific threat signatures within the broader attack categories:

- **Primary Categories:** The map is divided into three main color blocks representing **DDoS** (Light Blue), **Malware** (Orange), and **Intrusion** (Purple).
- **Sub-categories:** Each of these primary blocks is further subdivided to show the proportion of specific signatures, labeled as **Known Pattern A** and **Known Pattern B**, indicating the makeup of recognized threats within each main attack type.



The final dashboard delivers the deepest threat profiling. A Total Attacks Decomposition tree lets analysts break attacks down by Attack Type and Severity interactively, supported by % High Severity (33.5%) and Malware Detected (20,000) cards. The Severity Mix by Network Segment and Action Taken by Attack Type charts relate severity and response to network structure and threat class, while the Attack Type & Signature treemap and the Attack Type across Anomaly Bands ribbon expose signature and anomaly patterns. A final Attacks by Attack Type × Severity matrix consolidates the numbers.

■ Dashboard 6 - Security Alert Monitoring

% Alert Rate

Positioned at the top of the left-hand KPI rail, this single-value card reports the % Alert Rate - the share of all 40,000 incidents that raised an alert or warning (Alerts/Warnings = “Alert Triggered”), calculated as $\text{DIVIDE}([\text{Total Alerts Triggered}], [\text{Attack}]) \times 100$.

Value displayed: 49.83% (19,933 of 40,000 incidents triggered an alert).

Because almost exactly half of every incident generates an alert, the card confirms that the alerting layer is highly active and, on its own, cannot be used to prioritise - roughly one in two events would

surface. This makes the severity- and response-based filtering elsewhere on the page essential for separating genuine signal from noise.

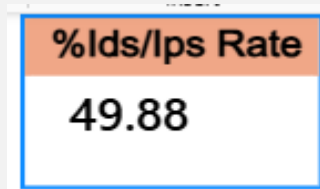


% IDS/IPS Rate

The second card on the KPI rail shows the % IDS/IPS Alert Rate - the proportion of incidents for which the intrusion-detection / intrusion-prevention system produced alert data (IDS/IPS Alerts = “Alert Data”), i.e. $\text{DIVIDE}([\text{IDS/IPS Alert Count}], [\text{Attack}]) \times 100$.

Value displayed: 49.88% (19,950 of 40,000 incidents carried an IDS/IPS alert).

The IDS/IPS layer fires on almost exactly half of all traffic, closely mirroring the general alert rate. The near-identical coverage suggests the two alerting mechanisms trigger largely in parallel rather than catching different populations of events - useful context when judging whether the sensors add independent value.



% Malware Detected

The third KPI card reports the % Malware Indicator Rate - the share of incidents flagged with a malware indicator of compromise (Malware Indicators = “IoC Detected”), $\text{DIVIDE}([\text{Malware Detected}], [\text{Attack}]) \times 100$.

Value displayed: 50.00% (exactly 20,000 of 40,000 incidents carry a malware IoC).

Malware indicators are present in precisely half of all records - the single largest risk signal on the page and consistent with the Milestone 3 finding of 20,000 IoC-positive incidents. It establishes that malware is not a niche threat in this environment but effectively a coin-flip on every event, which justifies the emphasis on blocking and firewall coverage across the rest of the dashboard.

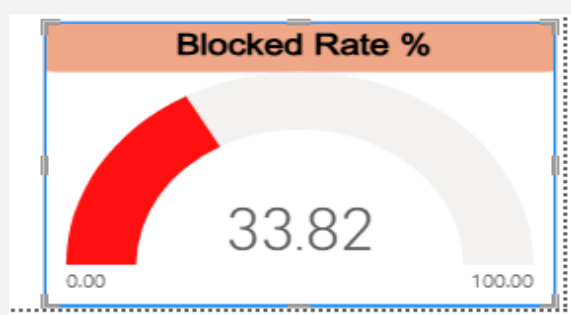


Blocked Rate %

Anchoring the foot of the KPI rail, this gauge visual plots the Blocked Rate % - the percentage of incidents that were actively Blocked (Action Taken = "Blocked"), $\text{DIVIDE}(\text{Blocked}, \text{Total}) \times 100$ - against a 0-100 scale.

Needle value: 33.82% (13,529 of 40,000 incidents blocked).

Only about a third of incidents are actually stopped at the point of detection, matching the 33.8% block rate reported in Milestone 3. Placing this gauge beside the ~50% alert and malware cards makes the central operational gap explicit: alerts fire on half of all events, yet only a third are blocked, so a large share of flagged - and potentially malicious - activity is still merely logged or ignored rather than prevented.



Alerts/Warnings Status by Attack Type

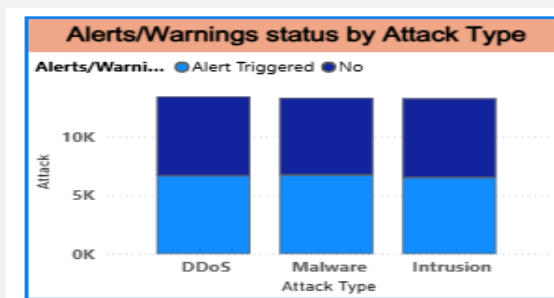
This clustered column chart sits at the top-centre of the page. It plots the Attack count on the Y-axis, grouped along the X-axis by Attack Type (DDoS, Intrusion, Malware) and split within each group by Alerts/Warnings status (Alert Triggered vs No).

DDoS - Alert Triggered: 6,673 · No alert: 6,755 (13,428 total)

Intrusion - Alert Triggered: 6,525 · No alert: 6,740 (13,265 total)

Malware - Alert Triggered: 6,735 · No alert: 6,572 (13,307 total)

Alert coverage is close to a 50/50 split within every attack type, with Malware the only class where alerts slightly outnumber non-alerts (6,735 vs 6,572). The near-uniformity shows the alerting system is essentially attack-type agnostic - it is not tuned to fire more aggressively on any one threat class - which is important when assessing whether high-risk malware is being surfaced often enough.

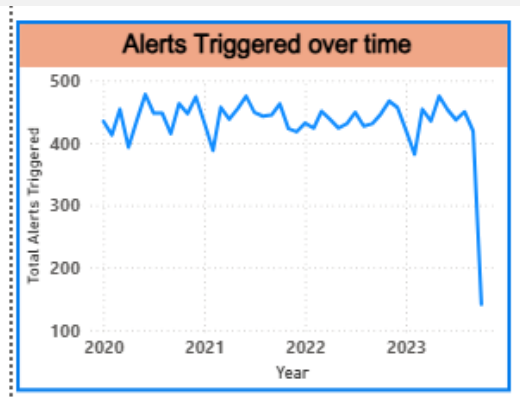


Alerts Triggered over Time

Running across the top-right, this line chart trends Total Alerts Triggered along a Year → Month date hierarchy, showing how alert volume has evolved from 2020 to 2023.

2020: 5,304 alerts · 2021: 5,286 · 2022: 5,277 · 2023: 4,066

Alert volume is essentially flat at roughly 5,300 per year across 2020-2022 before dropping to about 4,066 in 2023. As with the attack-volume trend seen in earlier dashboards, the 2023 decline reflects partial-year data rather than a genuine fall in threat activity, and the drill-down to month lets an analyst pinpoint exactly where coverage tapers off.



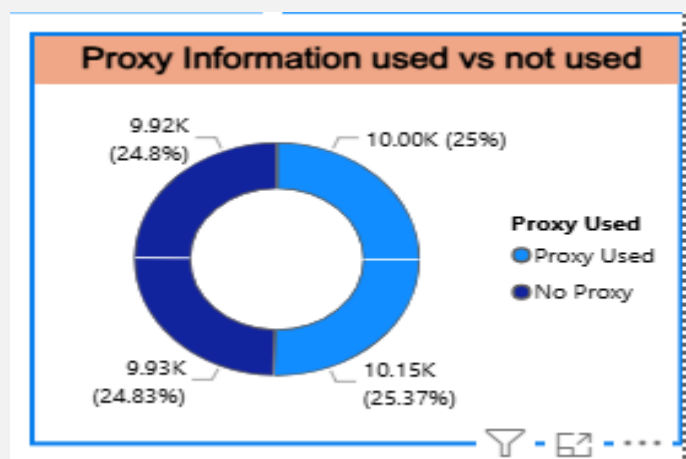
Proxy Information Used vs Not Used

This donut chart, lower-centre, breaks the total Attack count by whether a proxy was involved (Proxy Used vs No Proxy), with each arc further segmented by Alerts/Warnings status.

Proxy Used: 20,149 incidents (50.4%) - of which 10,000 triggered an alert

No Proxy: 19,851 incidents (49.6%) - of which 9,933 triggered an alert

Proxy usage is split almost evenly across the dataset, and the alert rate is virtually identical whether or not a proxy was used (~49-50% in both arcs). This tells the SOC that attackers' use of proxies is not materially changing whether an event gets alerted on - proxy presence is a useful forensic attribute but not, by itself, a reliable trigger for escalation.



Firewall Log Presence by Severity Level

The clustered bar chart at the lower-right measures Firewall Log Coverage (the count of incidents where Firewall Logs = “Log Data”) down each Severity Level (High, Medium, Low), split by whether a firewall log was present.

High severity - Log Data: 6,696 · No log: 6,686

Medium severity - Log Data: 6,757 · No log: 6,678

Low severity - Log Data: 6,586 · No log: 6,597

Total firewall-log coverage: 20,039 of 40,000 incidents (~50%).

Firewall logging covers almost exactly half of incidents at every severity level - critically, High-severity events are logged no more often (6,696 of 13,382) than Low-severity ones. The absence of severity-weighted logging is a genuine operational gap: the most dangerous incidents are just as likely to be missing firewall evidence as the least dangerous, with direct implications for forensic readiness.



Dashboard 6 shifts the analysis from what the attacks are to how well the monitoring and alerting stack is performing. Four headline KPIs - % Alert Rate (49.83%), % IDS/IPS Rate (49.88%), % Malware Detected (50.00%) and a Blocked Rate % gauge (33.82%) - frame a consistent story: detection signals fire on roughly half of all traffic, yet only a third of incidents are actively blocked. The surrounding visuals decompose that headline by dimension - alert status across attack types, alert volume over time, proxy involvement, and firewall-log coverage by severity - and together they expose where monitoring is strong (broad, even alert coverage) and where it is weak (blocking and severity-weighted firewall logging), giving the security operations centre a clear, evidence-based view of alert-handling effectiveness.

Dashboard 7 - Attack Trends & Response Effectiveness

Total Alert

This KPI card in the top-right reports Total Alerts Triggered - the raw count of incidents whose Alerts/Warnings field equals “Alert Triggered” (CALCULATE([Attack], Alerts/Warnings = “Alert Triggered”)).

Value displayed: 19,933 alerts.

Just under 20,000 of the 40,000 incidents raised an alert, providing the absolute baseline that the rest of this page evaluates responses against. It anchors the effectiveness story: with roughly 20k alerts in play, the central question the dashboard answers is how consistently those alerts convert into a “Blocked” outcome.



Attacks vs Blocked Over Time

Occupying the top-left, this dual-measure line chart trends two series on a Year → Month hierarchy: total Attack volume and the Blocked Rate, letting the reader see whether response effectiveness holds up as attack volume changes.

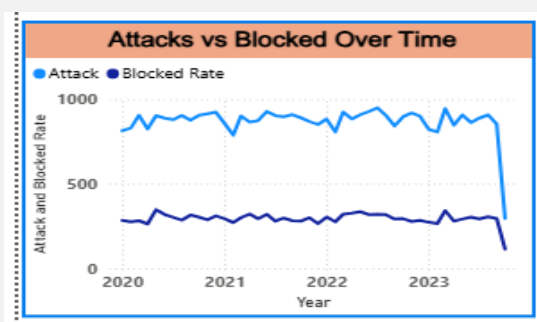
2020: 10,573 attacks · 33.86% blocked

2021: 10,538 attacks · 33.32% blocked

2022: 10,750 attacks · 34.16% blocked (peak year)

2023: 8,139 attacks · 33.98% blocked (partial year)

Attack volume is stable at ~10,500-10,750 per year through 2022 before the partial-2023 dip, while the block rate stays in a remarkably narrow 33.3-34.2% band throughout. That flat block-rate line is the key finding: response effectiveness has neither improved nor degraded over four years and appears independent of how many attacks arrive.



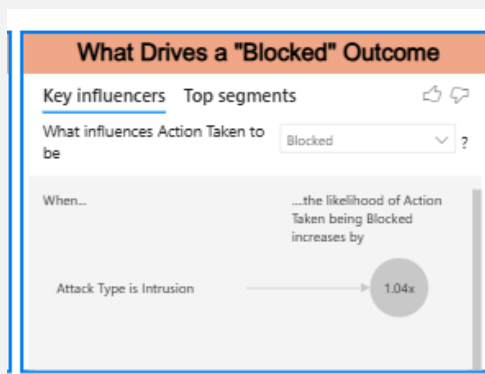
What Drives a “Blocked” Outcome

This Key Influencers visual, top-centre, uses Power BI’s built-in analytics to rank which factors most increase the likelihood that Action Taken = “Blocked”. The target is Action Taken and the model explains it using Attack Type, Severity Level, Protocol and Proxy Used.

Because the underlying categories are near-uniformly distributed (each attack type ≈33%, each severity ≈33%, block rate ≈33-34% everywhere), the visual surfaces only marginal influencers rather than a dominant driver.

The absence of a strong driver is itself the insight: no single attack type, severity, protocol or proxy state meaningfully changes the odds of an incident being blocked. This confirms that blocking in this

dataset is effectively uniform and not risk-weighted - a finding that would prompt a real SOC to introduce severity-aware blocking rules.



Response Summary by Attack Type

This table on the right-hand side summarises response performance per Attack Type, with columns for Attack (volume), Block Rate % and an Attack Dense Rank (RANKX over attack types by volume, dense/descending).

DDoS - 13,428 attacks · 33.76% blocked · rank 1

Malware - 13,307 attacks · 33.39% blocked · rank 2

Intrusion - 13,265 attacks · 34.32% blocked · rank 3

DDoS is the most frequent attack type (rank 1) yet has almost the lowest block rate (33.76%), while Intrusion - the least frequent - is blocked most often (34.32%). The spread is well under one percentage point, reinforcing that response is even-handed across threat classes rather than concentrated on the highest-volume one.

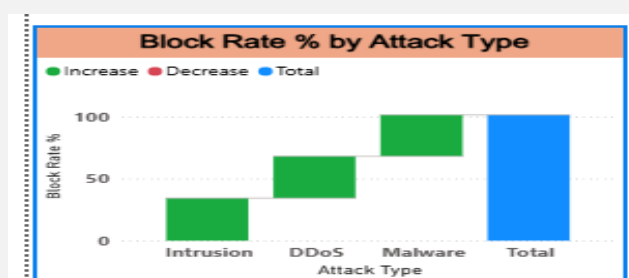
Response Summary by Attack Type			
Attack Type	Attack	Block Rate %	Attack Dense Rank
Intrusion	13265	34.32	3
Malware	13307	33.39	2
DDoS	13428	33.76	1

Block Rate % by Attack Type

This waterfall chart, bottom-left, decomposes Block Rate % across the three Attack Types, showing each type's contribution as a step and its cumulative position.

DDoS: 33.76% · Malware: 33.39% · Intrusion: 34.32%

The waterfall makes the near-equality visually explicit - the steps are almost the same height, so no attack type stands out as being blocked substantially more or less than the others. It is the graphical companion to the response table and drives home that block-rate parity, not variation, is the defining characteristic of the response profile.



Attack Type → Action Taken Flow

This Sankey diagram, bottom-centre, visualises the flow from each Attack Type (source) to each Action Taken (destination), with ribbon width weighted by Attack count - a single view of how every threat class is dispositioned into Blocked, Ignored or Logged.

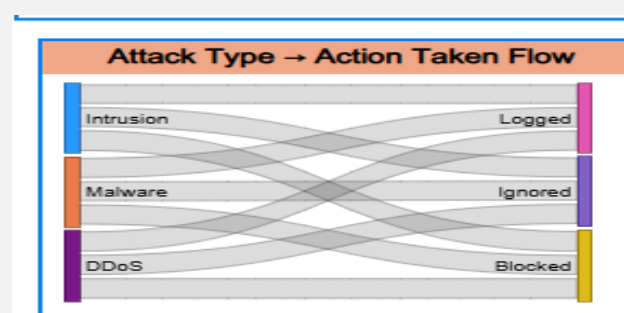
DDoS → Blocked 4,533 · Ignored 4,459 · Logged 4,436

Intrusion → Blocked 4,553 · Ignored 4,401 · Logged 4,311

Malware → Blocked 4,443 · Ignored 4,416 · Logged 4,448

Column totals: Blocked 13,529 · Ignored 13,276 · Logged 13,195

Every attack type splits into the three actions in almost equal thirds, so the ribbons are of near-identical width. The operationally important flows are the ~13,276 Ignored and ~13,195 Logged incidents: two-thirds of all activity - including roughly 8,900 malware-carrying events - is not being blocked, which is the single clearest efficiency concern the response analysis raises.

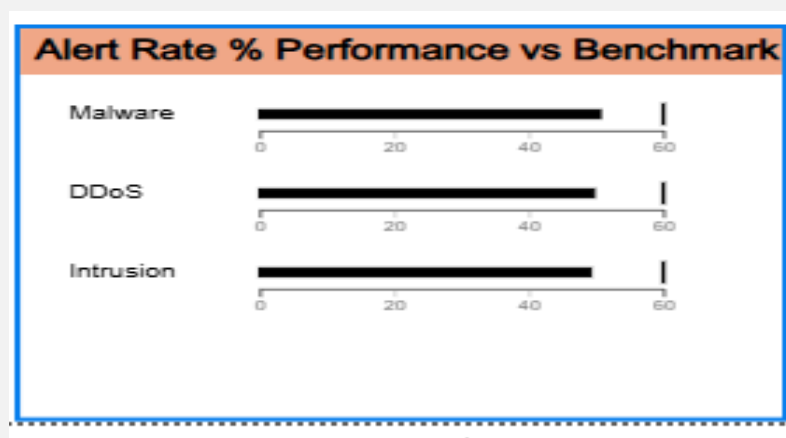


Alert Rate % Performance vs Benchmark

The bullet chart in the bottom-right compares each Attack Type's % Alert Rate against a common benchmark band, giving a compact "actual vs target" read on alerting performance by threat class.

DDoS: 49.69% · Intrusion: 49.19% · Malware: 50.61% (overall 49.83%)

All three attack types cluster tightly around the ~50% alert-rate benchmark, with Malware marginally ahead (50.61%) and Intrusion marginally behind (49.19%). The tight grouping confirms alerting is applied consistently regardless of threat type; the small Malware lead is the only positive signal that the most dangerous class is being surfaced slightly more often than average.



Dashboard 7 is the capstone of the framework, pivoting from monitoring coverage to response effectiveness and trend. A Total Alert card (19,933) sets the baseline, while the Attacks vs Blocked Over Time line shows attack volume peaking in 2022 (10,750) against a block rate that never leaves a narrow 33-34% band. The Key Influencers visual, the Response Summary table, the Block Rate % waterfall and the Attack Type → Action Taken Sankey all converge on the same conclusion: response is uniform - every attack type is blocked, ignored and logged in near-equal thirds, with no risk-weighting - while the Alert Rate bullet chart confirms alerting is applied evenly across threat classes. Taken together the page tells a clear operational story: the environment reliably detects and consistently responds, but blocks only about a third of incidents, leaving a two-thirds tail of ignored and logged activity that is the primary target for future tuning.

6. Snapshots / Screenshots

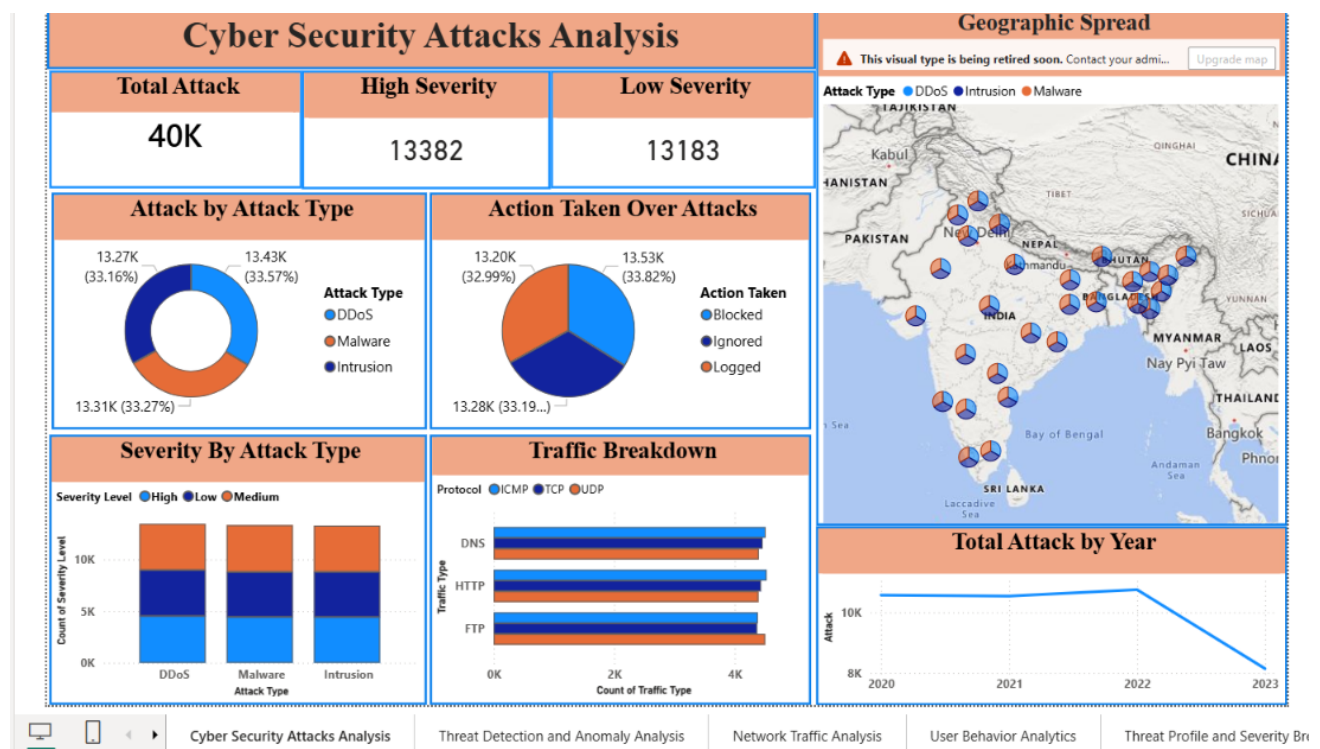
The interactive dashboard captures are embedded alongside their explanations in Section 5b as Pic 1.1 to Pic 1.7. The table below indexes each snapshot for quick reference. High-resolution screenshots exported directly from Power BI can be inserted in an appendix if a raw-capture set is required

Data Transformation Steps

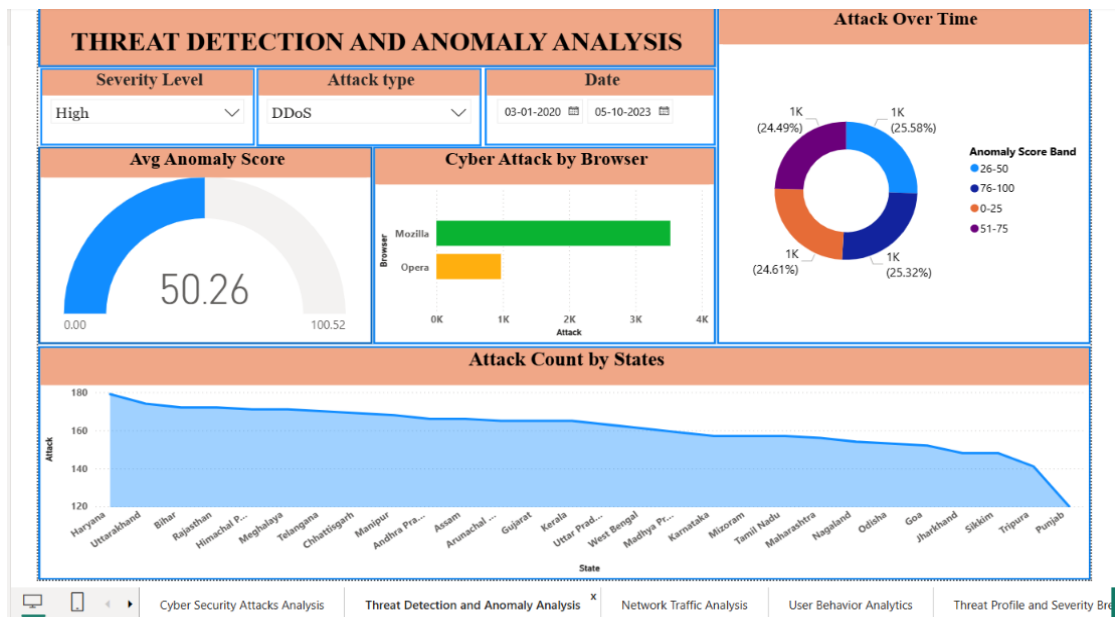
The screenshot shows the Power BI Desktop interface with the 'Transform' tab selected. The main view displays a table with 27 rows of cybersecurity attack data. The table has columns for 'Log Data', 'Alert Data', 'Log Source', 'Browser', 'Severity Sort', and 'Anomaly Score Band'. The 'Anomaly Score Band' column contains values like '1 26-50', '1 51-75', '1 76-100', '2 0-25', and '3 26-50'. The right-hand pane shows the 'Query Settings' for 'cybersecurity_attacks', including 'PROPERTIES' (Name: cybersecurity_attacks) and 'APPLIED STEPS' (Renamed Columns, Changed Type1, Replaced Value, Replaced Value1, Replaced Value2, Replaced Value3, Split Column by Delimiter1, Changed Type2, Renamed Columns1, Removed Columns, Changed Type3, Removed Duplicates, Removed Errors, Replaced Value4, Replaced Value5, Changed Type4, Replaced Value6, Renamed Columns2, Added Conditional Column, Changed Type5, Added Conditional Column1, Changed Type6, and Added Conditional Column2).

Line	Log Data	Alert Data	Log Source	Browser	Severity Sort	Anomaly Score Band
1	Log Data	No	Server	Mozilla		1 26-50
2	Log Data	No	Firewall	Mozilla		1 51-75
3	Log Data	Alert Data	Firewall	Mozilla		1 76-100
4	No	Alert Data	Firewall	Mozilla		2 0-25
5	No	Alert Data	Firewall	Mozilla		1 0-25
6	No	No	Server	Opera		2 0-25
7	Log Data	No	Server	Opera		3 26-50
8	Log Data	Alert Data	Firewall	Mozilla		3 51-75
9	Log Data	Alert Data	Server	Mozilla		3 51-75
10	No	No	Server	Mozilla		2 0-25
11	No	No	Server	Mozilla		2 0-25
12	No	No	Firewall	Opera		1 76-100
13	Log Data	Alert Data	Server	Opera		2 51-75
14	No	No	Firewall	Mozilla		1 51-75
15	No	No	Firewall	Mozilla		3 51-75
16	Log Data	No	Firewall	Opera		2 76-100
17	Log Data	No	Server	Opera		3 26-50
18	Log Data	Alert Data	Firewall	Mozilla		1 26-50
19	No	No	Firewall	Mozilla		3 26-50
20	Log Data	Alert Data	Server	Mozilla		3 0-25
21	Log Data	No	Firewall	Mozilla		2 76-100
22	Log Data	Alert Data	Server	Mozilla		1 76-100
23	No	Alert Data	Server	Mozilla		3 51-75
24	Log Data	No	Server	Mozilla		2 51-75
25	Log Data	Alert Data	Firewall	Mozilla		3 76-100
26	No	Alert Data	Firewall	Mozilla		1 26-50
27	Log Data	Alert Data	Firewall	Opera		1 26-50

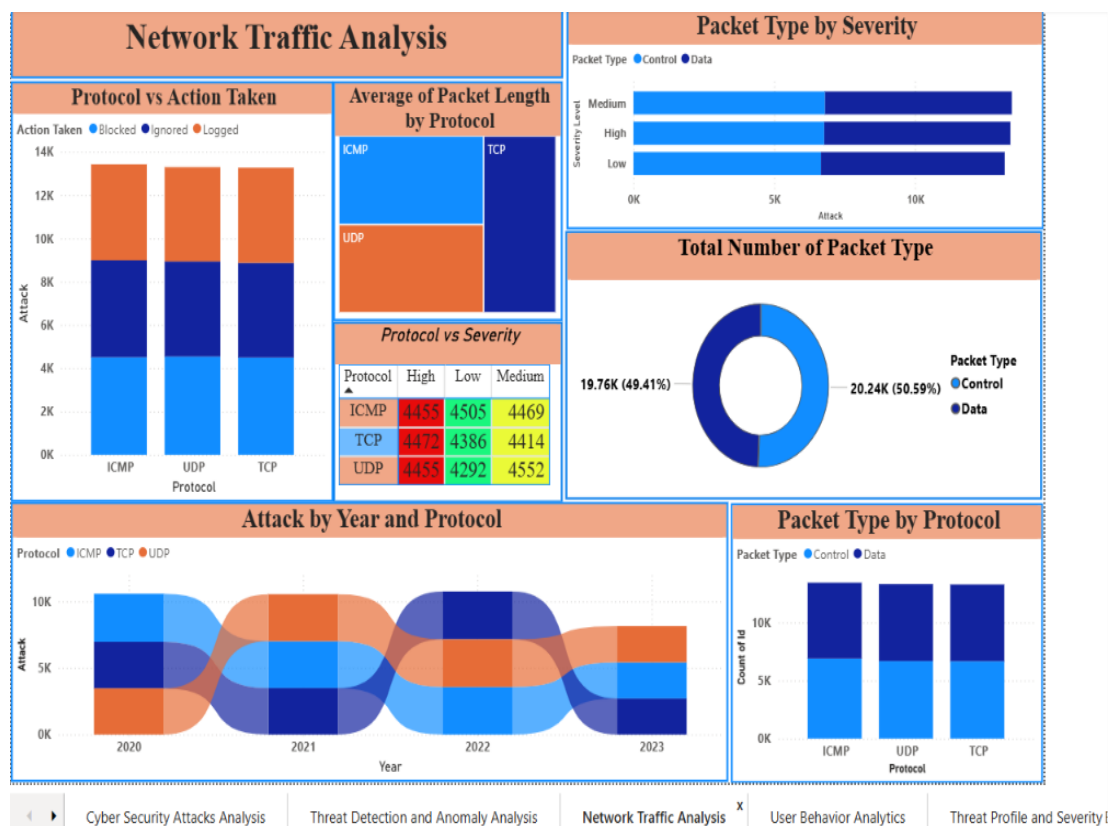
Dashboard 1 - Cyber Security Attacks Analysis



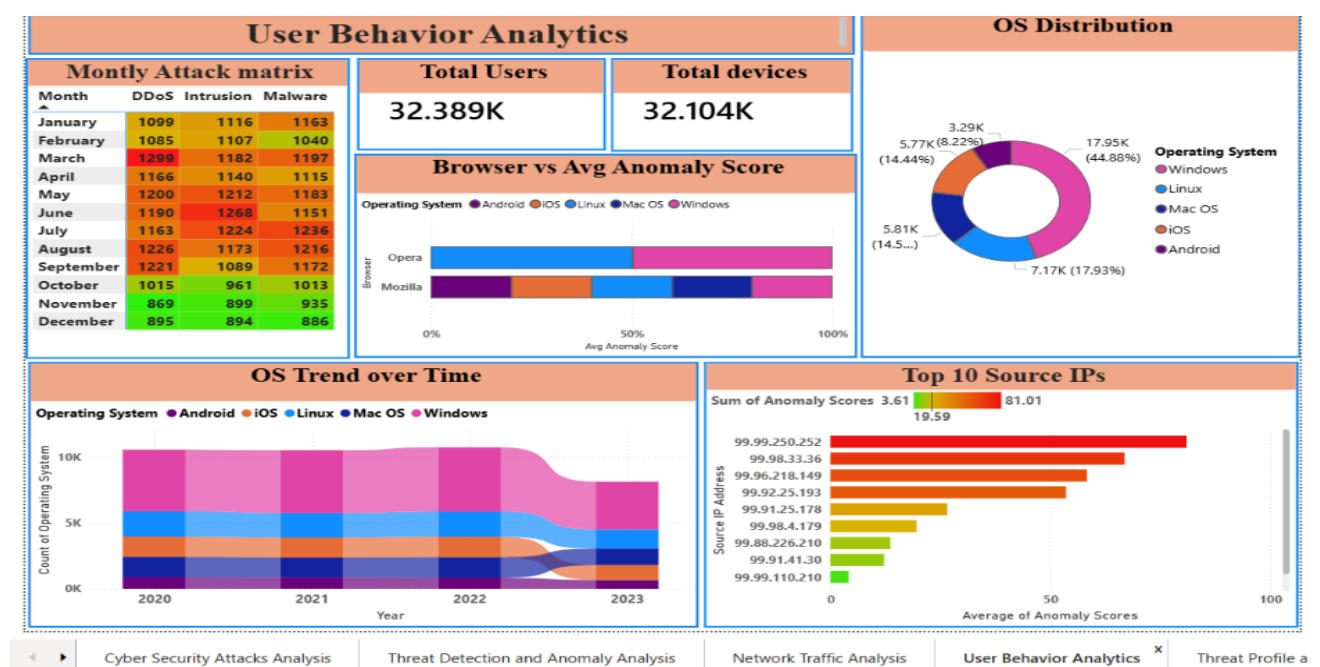
Dashboard 2 - Threat Detection and Anomaly Analysis



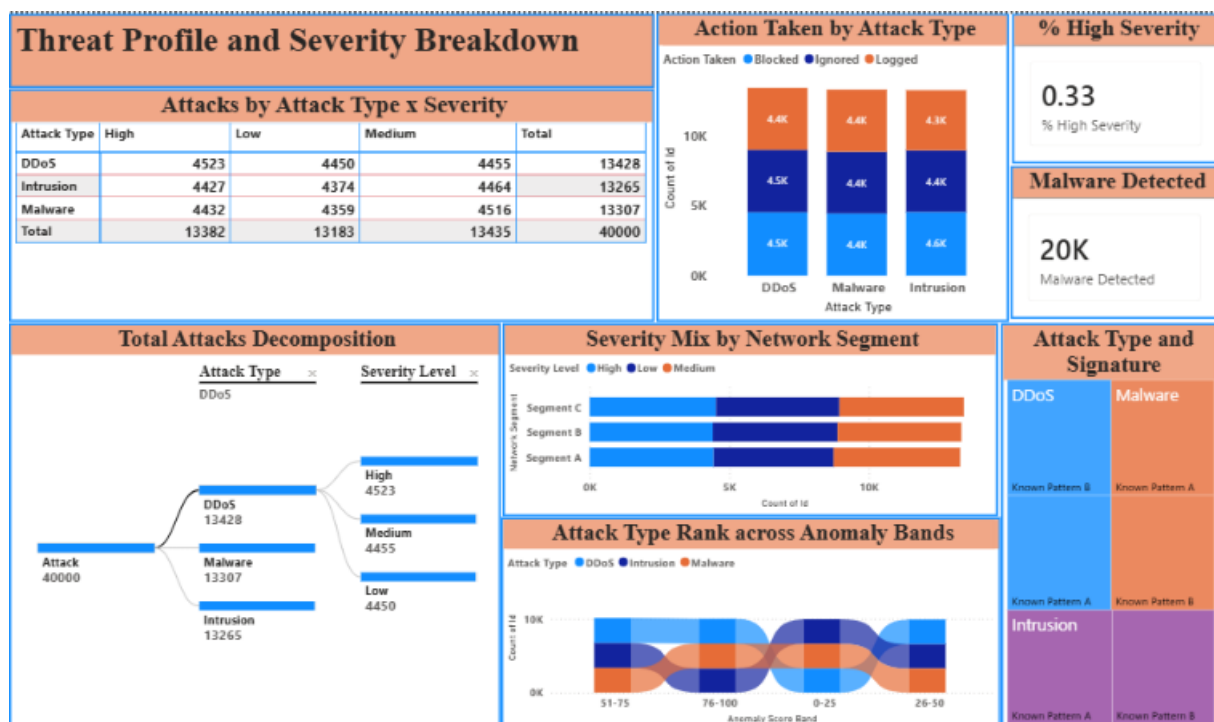
Dashboard 3 - Network Traffic Analysis



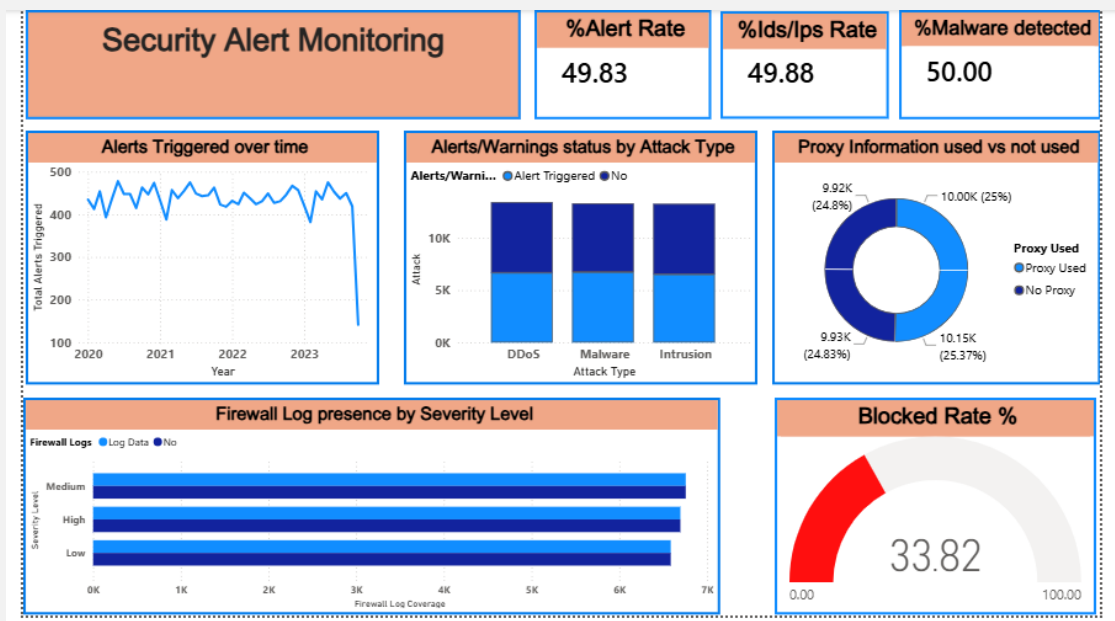
Dashboard 4 - User Behaviour Analytics



Dashboard 5 - Threat Profile and Severity Breakdown



Dashboard 6 - Security Alert Monitoring



Dashboard 7 - Attack Trends & Response Effectiveness



7. Challenges Faced

The advanced-analytics and reporting phase of Milestone 3 presented several challenges, each of which was resolved through iteration and collaboration:

- Handling a 40,000-row model while keeping visuals responsive - this required efficient DAX and disabling unnecessary auto date/time tables.
- Designing meaningful DAX for derived metrics (% High Severity, Block Rate %, Anomaly Score Band bucketing) so that KPIs stayed accurate under all slicer interactions and filter contexts.
- Configuring the decomposition tree and ribbon charts to reveal hierarchy (Attack Type → Severity) without overwhelming the reader.
- Maintaining a consistent visual language - colours, fonts and severity encoding - across seven dashboards contributed to by different members.
- Representing geography meaningfully across 28 states without producing a cluttered or misleading map.
- Balancing dashboard density - showing all key visuals on each page while preserving readability.
- Coordinating remotely via Microsoft Teams and merging individual work through GitHub with minimal conflicts.
- Interpreting a near-uniformly distributed dataset - deriving genuine, defensible insights rather than over-reading statistical noise.

8. Learnings & Skills Acquired

■ 8.1 Technical Skills

- End-to-end Power BI Desktop: data import, Power Query ETL, modeling, visualization and publishing.
- Writing DAX measures and calculated columns (CALCULATE, DIVIDE, COUNTROWS, AVERAGE and conditional banding).
- Building a date table and managing relationships for time-based analysis.
- Using advanced visuals: decomposition tree, ribbon chart, treemap, gauge, matrix, KPI cards and maps.
- Report theming, slicers, cross-filtering and drill-through.
- Version control with GitHub for the .pbix and documentation artefacts.

■ 8.2 Analytical and Problem-Solving Skills

- Translating a business/security question into measurable KPIs.
- Selecting the most appropriate visual for each analytical intent.
- Reading distributions, proportions and trends to form defensible conclusions.
- Debugging DAX and filter-context issues.
- Validating numbers across visuals to ensure consistency.

■ 8.3 Future Enhancements

- Integrate near-real-time data via scheduled refresh or DirectQuery.
- Add predictive analytics (anomaly forecasting, risk scoring) using Python/R or Azure ML.
- Implement row-level security for role-based dashboards (SOC analyst vs manager).
- Automate alerting on threshold breaches with Power Automate.
- Enable drill-through to raw incident detail and add a true India choropleth map.
- Publish to the Power BI Service within an app workspace for wider stakeholder access.

■ 8.4 Soft Skills and Team Collaboration

- Task distribution, ownership and accountability within the team.
- Communicating analytical findings clearly to non-technical audiences.
- Structured collaboration and a regular review cadence via Microsoft Teams.
- Time management against milestone deadlines.
- Giving and receiving constructive feedback and iterating on it.

■ 8.5 Domain Knowledge and Application

- Core cybersecurity concepts: attack types (DDoS, Malware, Intrusion), severity levels, protocols, packet types, malware indicators (IoC), anomaly scoring, network segments and response actions (Blocked / Ignored / Logged).
- How security operations teams triage and respond to incidents.
- How business intelligence supports security monitoring, reporting and executive decision-making.

9. Testimonials from Team

Siddhant

Siddhant played a pivotal role throughout this project and particularly during Milestone 3. He led the data-modeling and DAX layer of the framework (Steps 4 and 5), building and marking the Date table, establishing the model relationships, and authoring the key measures - including % High Severity, Block Rate %, Malware Detected and the Anomaly Score Band bucketing logic - that power the KPIs across every dashboard. He also designed the Threat Profile and Severity Breakdown dashboard, configuring the decomposition tree and the severity matrices that give the report its analytical depth. Beyond his technical contribution, Siddhant coordinated task allocation on Microsoft Teams, reviewed team members' visuals to keep the colour and severity encoding consistent, resolved GitHub merge conflicts when individual work was combined, and patiently helped his peers understand filter context and visual selection. His ownership, clear communication and willingness to support the team were instrumental in delivering this milestone on time..

Priyam Sinha

Priyam Sinha played a key role in framing the project and translating its analysis into a finished, communicable product. He led the problem-definition stage (Step 1), helping shape the security questions and KPIs the framework needed to answer, and later drove the insight-generation and reporting stages (Steps 8 and 9). Priyam built the Threat Detection and Anomaly Analysis and User Behaviour Analytics dashboards - designing the anomaly-score gauge, the anomaly-band and OS-distribution visuals, and the top source-IP and user/device views - and worked closely with the team to interpret the results and validate figures across dashboards for consistency. He also took ownership of the documentation effort, consolidating findings into the project report and PowerPoint walkthrough and maintaining the GitHub repository. Priyam's structured thinking, strong communication and eye for presentation gave the final deliverables their clarity and polish.

Ponnuru Nithin

Ponnuru Nithin was central to the data-engineering backbone of the project, owning the data-collection and preparation stages (Steps 2 and 3). He sourced and validated the 40,000-record cybersecurity dataset from Kaggle, then shaped it in the Power Query Editor - correcting data types, handling null and inconsistent values, splitting the date and time fields, standardising categorical values such as protocol, packet type and action taken, and adding the Severity Sort column that controls ordering across the visuals. Building on this clean foundation, he designed and built the Network Traffic Analysis dashboard, crafting the packet-type, protocol and severity visuals along with the Protocol vs Severity matrix. Nithin also contributed to overall dashboard interactivity (Step 7) by configuring slicers

and cross-filtering, and was quick to troubleshoot data-refresh and query issues for the rest of the team. His attention to data quality and his reliability under deadline pressure ensured that every downstream analysis rested on trustworthy, well-structured data.

10. Conclusion

This project successfully transformed a large, raw cybersecurity dataset of 40,000 incidents into an Advanced Threat Incident Analytics and Security Visualization Framework comprising seven interactive Power BI dashboards. From attack overview to anomaly detection, network traffic, user behaviour and deep threat profiling, the framework provides a complete, drill-able view of the threat landscape and converts complex log data into intelligence that both technical and non-technical audiences can act on.

The analysis showed a broad, multi-vector threat environment: attack types and severity levels are almost evenly distributed, Windows is the most-targeted operating system at around 45%, malware indicators appear in half of all events, and only about a third of attacks are actively blocked - a clear, quantified opportunity to strengthen response. More importantly, the project reinforced the value of data-driven decision-making and gave the team hands-on mastery of Power BI, DAX, data modelling and analytical storytelling, alongside the collaboration and communication skills essential to professional project delivery. Milestone 3 marks a meaningful step in the team's academic and professional journey and a strong foundation for the future enhancements outlined in Section 8.3.

11. Acknowledgements

We would like to express our sincere gratitude to the Infosys Springboard Team for organizing this highly enriching virtual internship program and for providing a structured, industry-aligned learning platform. The well-designed modules, milestone-based framework, and practical orientation of the internship enabled us to gain meaningful exposure to real-world business intelligence applications.

We extend our heartfelt appreciation to our mentor, Ms. Nityasree, for her consistent guidance, valuable feedback, and continuous encouragement throughout the project lifecycle. Her insights and structured direction played a significant role in shaping the analytical depth and professional quality of this project. We would also like to acknowledge our team members for their strong collaboration, accountability, and shared commitment toward achieving project objectives. The coordinated distribution of responsibilities, open communication, and collective problem-solving approach significantly contributed to the successful execution of this project.

In addition, we extend our appreciation to the creators and maintainers of Microsoft Power BI documentation and community resources, which served as essential references during dashboard development and KPI implementation. We are also grateful to Kaggle for providing access to publicly available datasets that facilitated practical, hands-on analytical experience.

The structured learning resources and practical exposure provided through this internship have significantly strengthened our analytical thinking, technical proficiency, and business interpretation skills. This experience has enhanced our ability to approach data with a strategic mindset, collaborate effectively within a team environment, and execute projects with professional discipline. Overall, this internship has been a meaningful milestone in our academic and professional journey, reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.